

K-FIELD MASS BRIDGE DRIVES VOLUME-FILLING HIERARCHY

A foundational textbook reference on the locked Cell3 shape, Lucas Group aggregation, the unique $9 \times 9 \times 9$ Mass-Bridge completion block, direct and reciprocal scaling, and harmonic tests against established physics

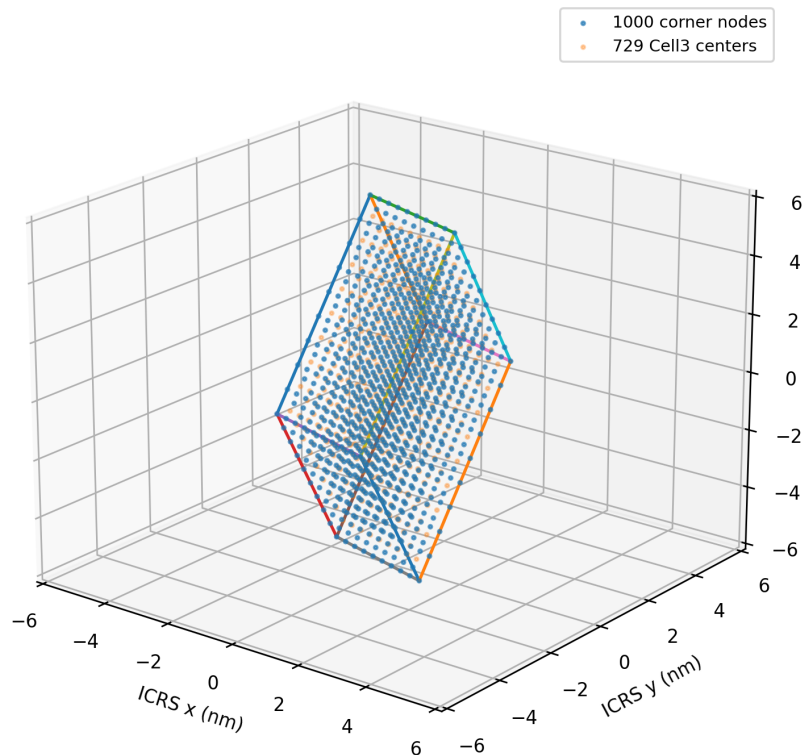
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Mass-Bridge completion block: $9 \times 9 \times 9$ Cell3 volumes



Title figure. Dimensionally correct ICRS rendering of the $g=2$ Mass-Bridge completion block. The $9 \times 9 \times 9$ space-filling arrangement contains 729 Cell3 centers and 1,000 merged unrefined corner nodes. Every plotted coordinate is generated from the locked Cell3 basis B.

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Document Control

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This volume is self-contained. It begins with the exact locked Cell3 geometry, derives the space-filling hierarchy without changing shape, identifies the unique aggregation rung selected by the Mass Bridge, develops the reciprocal spectrum, and separates exact identities from empirical harmonic comparisons.

Contents

Document Control	2
Abstract	6
1. Reader Orientation and Claim Structure	6
2. Locked Mass-Bridge Geometry	6
2.1 The projective face-area point	6
2.2 Uniform scale is the only free degree	7
2.3 Authoritative Cell3 numerical kernel	8
2.4 Direct and reciprocal metrics	9
3. Cell3 Nodes and Boundary Planes	10
3.1 Exact 27-node rule	10
3.2 Six boundary planes	10
4. Space-Filling Construction of the Lucas Group	11
4.1 Three complete cells per primitive direction	11
4.2 The two different meanings of 27	12
4.3 Collective phase sectors and internal connectivity	12
5. The Second Aggregation and the Mass-Bridge Completion Theorem	13
5.1 Aggregation depth	13
5.2 The 10-100-1000 node skeleton	13
5.3 Uniqueness of the selected rung	14
5.4 Exact commutative closure	14
5.5 Full g=2 inventory	15
6. Accumulated Base Dimensions	16
6.1 Direct scaling	16
6.2 The complete g=2 dimensions	17
7. General Volume-Filling Hierarchy	17
7.1 Closed-form count laws	17
7.2 Interfaces, exposed faces, and graph cycles	18
7.3 Complete hierarchy table	18
7.4 Exact sheet-to-volume recursion	18
7.5 Parity conservation and residual one	19
8. From Local Object to Collective Field Volume	20

8.1 $g=0$: local geometric object	20
8.2 $g=1$: first collective resonator	20
8.3 $g=2$: Mass-Bridge-complete finite object	20
8.4 $g=3$: bulk-dominated field volume	20
8.5 Interpretation of increasing complexity	21
9. Reciprocal Space and Harmonic Families	21
9.1 Allowed reciprocal vectors	21
9.2 Direct and reciprocal scaling table	22
9.3 Enumerated low-order reciprocal spectrum	22
9.4 Collective phase sectors	23
10. Harmonics and Subharmonics of Established Physics	23
10.1 Controlled comparison method	23
10.2 Strongest retained candidates	24
10.3 Statistical reading of the 864 candidate	24
10.4 Nuclear-count candidates at the Mass-Bridge level	25
10.5 Pion and tau structural counts	25
10.6 Why unrestricted matching is rejected	25
11. Testable Consequences and Research Program	25
11.1 Minimal reciprocal-spectrum test	25
11.2 Boundary-to-bulk transition tests	25
11.3 Sheet inheritance	25
11.4 Residual recursion	25
12. Principal Findings	26
13. Conclusions	26
References and Source Record	26
Appendix A. Complete Locked Basis, Metrics, and Planes	27
Appendix B. Complete 27-Node Ledger	28
Appendix C. Hierarchy Ledger through $g=6$	28
Appendix D. Low-Order Reciprocal Quadratic Spectrum	29
Appendix E. Complete 26-Quantity Comparison Ledger	30
Appendix F. Complete Consolidated Analytical Statement	30
Principal Result: The Mass Bridge Selects The Second Lucas Aggregation	30

Accumulated Physical Dimensions 31

Growth Of Internal Structure 31

Object-To-Field Transition 31

Reciprocal-Space Harmonics 31

Comparison With Established Physics Constants 32

Finding 32

Appendix G. Reproducibility Equations **32**

Abstract

The Mass Bridge fixes the Cell3 face-area ratios to the proton-electron mass ratio, the inverse fine-structure constant, and their product. Once these ratios are locked, Cell3 is no longer a freely adjustable shape: only one common positive scale remains. This changes the natural analysis. Instead of fitting an unconstrained cell to external quantities, one can construct exact space-filling copies of the locked cell and ask which collective counts, dimensions, reciprocal modes, and residual structures arise at each aggregation level.

A Lucas Group contains three complete Cell3 volumes along each primitive direction, for 27 volumes and outer basis 3B. Repeating the same space-filling operation produces aggregation depth g with 3^g Cell3 units per axis. The second aggregation, $g=2$, contains nine Cell3 units per axis. Its unrefined boundary lattice has ten nodes per axis, one hundred nodes per face, and one thousand nodes per volume. Those counts are exactly the 10, 100, and 1000 multipliers in the locked projective Mass Bridge. The equation $3^g 2^n + 1 = 10$ has only the nonnegative solution $g=2, n=0$, so the Mass Bridge selects a unique unrefined recursive rung.

The resulting 9B object contains 27 Lucas Groups, 729 Cell3 volumes, 1,000 corner nodes, 6,859 merged half-step nodes, 55 constant-lambda sheets, 3,429 antipodal node pairs, and 729 cell-center phase sectors. It is the smallest hierarchy volume in which the complete one-dimensional, two-dimensional, and three-dimensional Mass-Bridge multiplication law is represented by the lattice itself.

The direct hierarchy scales by powers of three, while internal midpoint refinement scales by powers of two. Their reciprocal actions are inverse: aggregation creates subharmonics 3^{-g} and midpoint refinement creates harmonics 2^n . The combined linear-dispersion family is $2^n/3^g$; the corresponding quadratic family is $4^n/9^g$. Low-order reciprocal modes are enumerated directly from the locked reciprocal metric. A controlled comparison with established dimensionless constants retains several residual candidates, led by the electron-to-shielded-helion magnetic-moment ratio near $864=2^5 3^3$, while multiple-testing correction keeps those matches in the candidate, rather than completed-law, category.

1. Reader Orientation and Claim Structure

Four kinds of statement appear in this paper. Exact source statements are values and rules carried directly from the locked Mass-Bridge and Lucas hierarchy files. Exact derived statements follow algebraically from those values and rules. Structural interpretations explain what the exact identities mean for a space-filling hierarchy. Empirical comparisons place the resulting integer and harmonic families beside measured physical quantities and inspect the residuals. Keeping these classes separate allows a student to reproduce every exact result before considering broader physical interpretation.

Class	Meaning	Examples in this paper
Locked source	Directly specified by the authoritative files	Basis B, direct metric M, reciprocal metric R, face ratios, 27-node rule
Exact derivation	Follows without fitted coefficients	$g=2$ uniqueness, 10/100/1000 closure, node and sheet formulas
Structural interpretation	Meaning assigned to exact hierarchy behavior	object-to-field transition, sheet-to-volume recursion
Empirical comparison	Measured quantity compared with a predetermined hierarchy candidate	864.058255 versus 864; mass-ratio residuals
Forward test	Calculation that can decide or refine the framework	complete $q^T R q$ spectral comparisons with one global scale

The central result is exact and independent of any particle-spectrum fit: the locked Mass Bridge selects the second Lucas aggregation because that level alone produces the required 10, 100, and 1000 node cardinalities in one, two, and three dimensions.

2. Locked Mass-Bridge Geometry

2.1 The projective face-area point

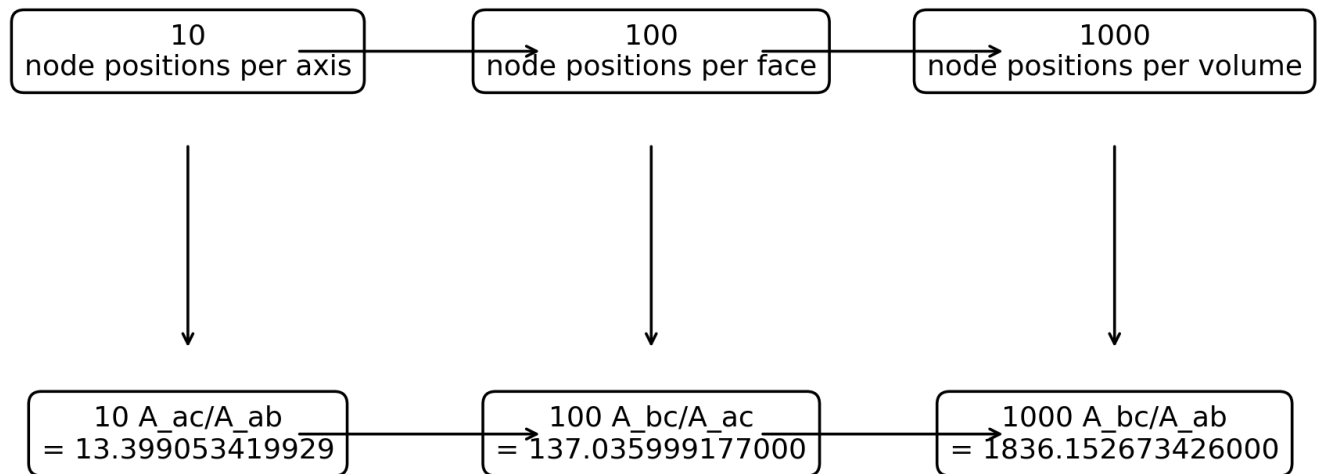
Let A_{ab} , A_{ac} , and A_{bc} be the areas of the three primitive faces of Cell3. The locked source imposes three dimensionless conditions:

$$\begin{aligned}
 1000 A_{bc}/A_{ab} &= 1836.152673426000 = \eta \\
 100 A_{bc}/A_{ac} &= 137.035999177000 = \alpha^{-1} \\
 10 A_{ac}/A_{ab} &= 13.399053419929 = \eta \alpha
 \end{aligned}$$

Only two face ratios are algebraically independent because the third closes by multiplication. The three equations are nevertheless useful because they expose three dimensional roles: a factor of ten attached to a line count, one hundred attached to a face count, and one thousand attached to a volume count.

Quantity	Exact locked value	Role
A_{ac}/A_{ab}	1.339905341992922	First independent shape ratio
A_{bc}/A_{ac}	1.370359991770000	Second independent shape ratio
A_{bc}/A_{ab}	1.836152673426000	Closure product $r_1 r_2$
$10 A_{ac}/A_{ab}$	13.399053419929221	$\eta \alpha$
$100 A_{bc}/A_{ac}$	137.035999177000036	α^{-1}
$1000 A_{bc}/A_{ab}$	1836.152673426000092	η

Mass-Bridge dimensional closure selected by the $g=2$ node skeleton



Exact closure: $(10 r_1)(100 r_2) = 1000(r_1 r_2)$, so $(\eta \alpha)(\alpha^{-1}) = \eta$

Figure 2.1. Exact dimensional closure. At $g=2$, ten lattice positions lie along each axis, one hundred on each unrefined face, and one thousand in the unrefined corner-node volume. Multiplying the corresponding locked face ratios reproduces the Mass-Bridge constants without adding a scale parameter.

2.2 Uniform scale is the only free degree

The Cell3 shape includes edge ratios, included angles, face ratios, handedness, and reciprocal anisotropy. Multiplying all three basis vectors by a common positive scalar s changes size but leaves the complete shape address unchanged.

```

B(s) = s B_locked
edge(s) = s edge(1)
area(s) = s^2 area(1)
volume(s) = s^3 volume(1)
reciprocal basis(s) = s^-1 reciprocal basis(1)

```

This distinction is essential. Absolute dimensions written in nanometres refer to the adopted realization in the locked source. Every ratio, count, angle, normalized projection, and reciprocal-direction relation survives any common rescaling.

2.3 Authoritative Cell3 numerical kernel

Measure	Value
a	0.334179679059111 nm
b	0.586410890411141 nm
c	0.792319765042685 nm
theta_ab	83.933483012313 deg
theta_ac	80.448130018338 deg
theta_bc	50.363231813756 deg
A_ab	0.194869165460484 nm^2
A_ac	0.261106235790205 nm^2
A_bc	0.357809539128562 nm^2
Volume	0.117914891911745 nm^3
Longest body diagonal	1.343653621675925 nm
Handedness	Left-handed ordered basis, det(B) < 0

B_nm =

[[0.327501242055891, 0.035330775843828, -0.010129317626729],

[-0.054897007884006, 0.207319850440192, -0.389787702366260],

[-0.037487502894730, -0.547401085388508, -0.689734408367268]]

Locked Cell3: 27-node midpoint grammar and authoritative edge basis

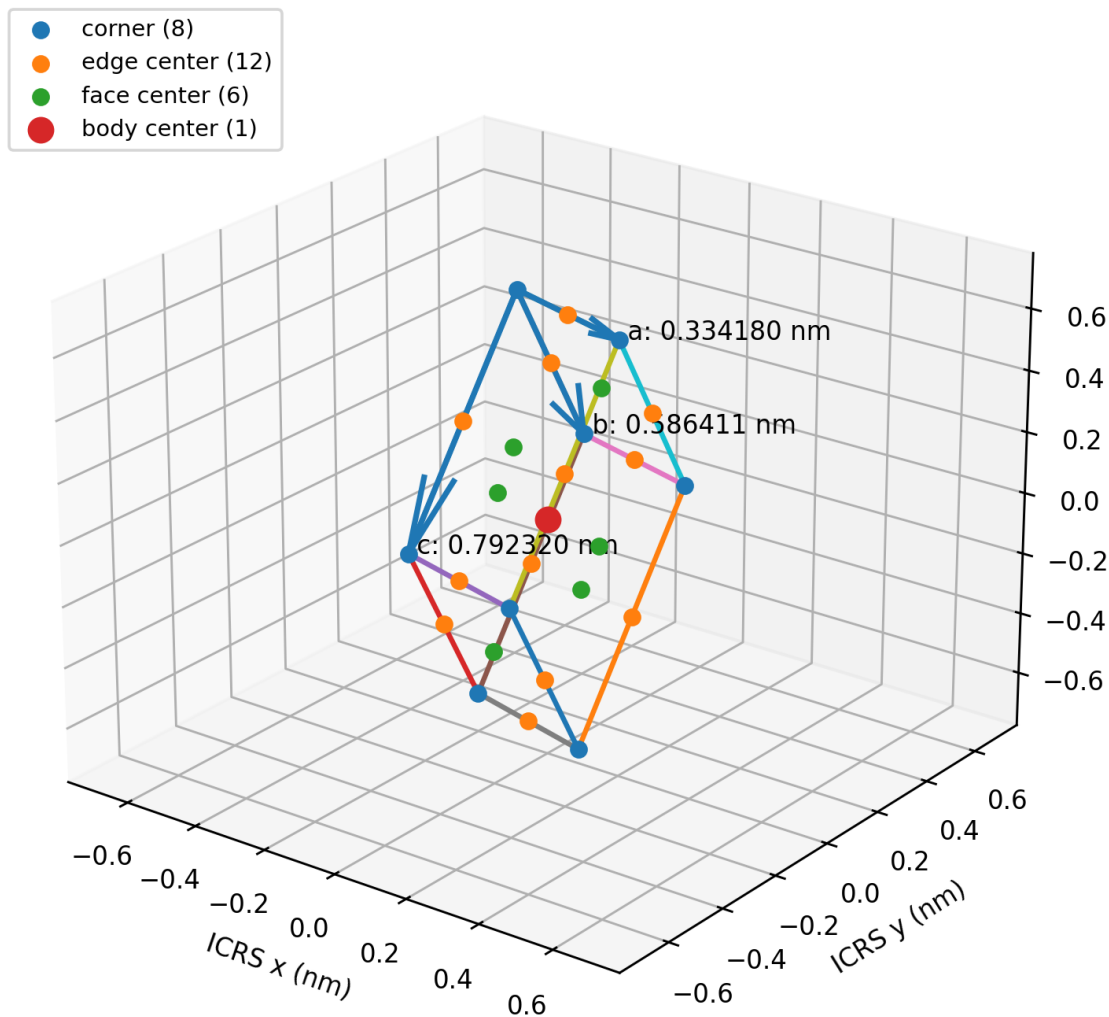


Figure 2.2. Dimensionally correct locked Cell3 in ICRS coordinates. The plot shows the full 27-node midpoint grammar, the enclosing parallelepiped, and the three ordered basis vectors with measured edge lengths.

2.4 Direct and reciprocal metrics

The direct metric M stores every squared length and inner product in primitive coordinates. The reciprocal metric $R=M^{-1}$ stores the corresponding plane-normal and wave-vector quadratic form. They are not independent fitted matrices; both are determined by B .

$$\begin{aligned}
 M &= B^T B \\
 R &= M^{-1} = B^{-1} B^{-T} \\
 ||B u||^2 &= u^T M u \\
 ||B^{-T} q||^2 &= q^T R q
 \end{aligned}$$

Matrix	Row 1	Row 2	Row 3
$M \text{ (nm}^2\text{)}$	0.111676058, 0.020710333, 0.043937235	0.020710333, 0.343877732, 0.296392759	0.043937235, 0.296392759, 0.627770610
$R \text{ (nm}^{-2}\text{)}$	9.208025529, 0.001534802, -0.645188036	0.001534802, 4.903398312, -2.315175584	-0.645188036, -2.315175584, 2.731171275

3. Cell3 Nodes and Boundary Planes

3.1 Exact 27-node rule

Each node is indexed by an address $L=(L_a, L_b, L_c)$ with every component in $\{-1, 0, 1\}$. Its physical position is

$$p_L = B L / 2, \quad L \text{ in } \{-1, 0, 1\}^3.$$

The 27 combinations consist of eight corners, twelve edge centers, six face centers, and one body center. The integer lambda address is $\text{lambda}=L_a+L_b+L_c$. The polarity used by the hierarchy is $(-1)^\lambda$, with the locked source recording the equivalent alternating sign at every node.

Node class	Address condition	Count
Corner	All three components nonzero	8
Edge center	Exactly two components nonzero	12
Face center	Exactly one component nonzero	6
Body center	$L=(0,0,0)$	1
Total	All addresses	27

Complete Cell3 address atlas: all 27 nodes, lambda values, and polarities

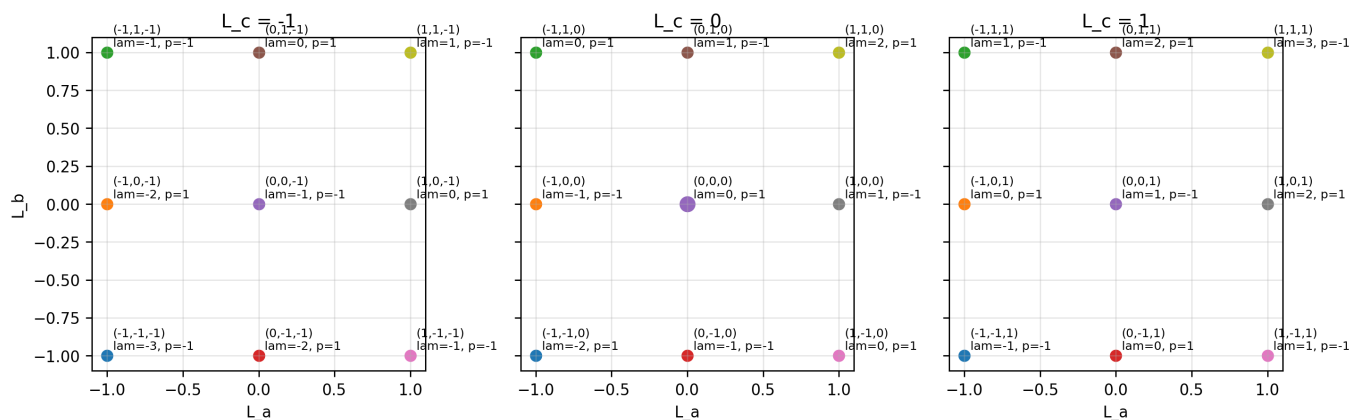


Figure 3.1. Complete address atlas. Every one of the 27 nodes is explicitly labeled by its integer address, lambda sheet, and polarity. The panels are exact coordinate slices, not a schematic.

3.2 Six boundary planes

A Cell3 has two boundary planes in each primitive channel. Every boundary face contains nine of the 27 midpoint nodes. The face normal is fixed by the reciprocal basis, while the opposite-face spacing is the physical thickness of one Cell3 in that channel.

Channel	Face family	Unit normal in ICRS	Face area (nm ²)	Opposite spacing (nm)
a	bc	-0.995965190, 0.083602163, -0.032619291	0.357809539129	0.329546529695
b	ac	-0.089052594, -0.866576769, 0.491033948	0.261106235790	0.451597379721
c	ab	0.194092201, 0.913177169, 0.358379235	0.194869165460	0.605097741519

Authoritative Cell3 boundary-plane atlas: six faces, nine nodes per face

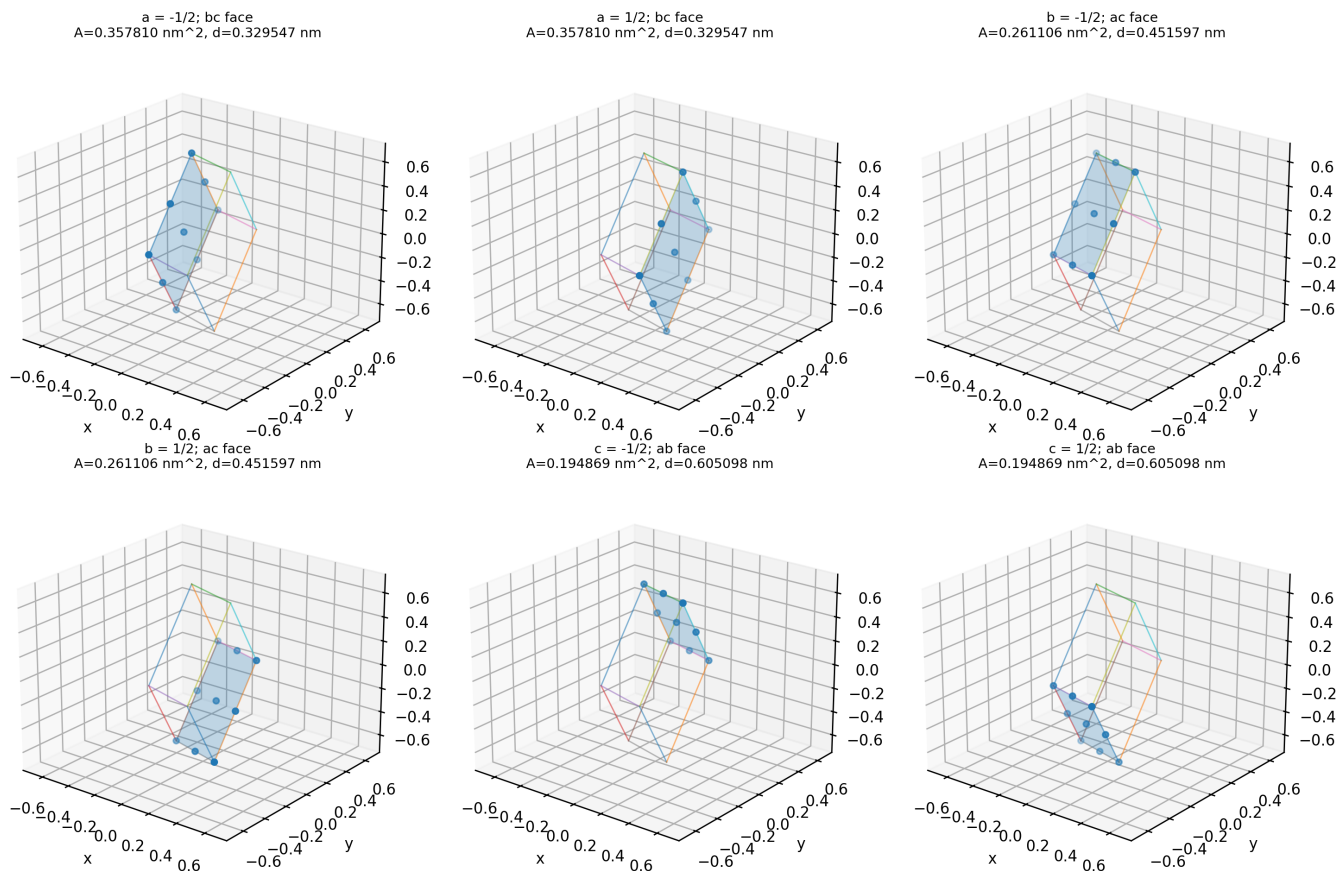


Figure 3.2. Six-plane atlas computed from the locked basis. Each panel shows one boundary face, its nine midpoint nodes, its measured area, and its opposite-face spacing.

4. Space-Filling Construction of the Lucas Group

4.1 Three complete cells per primitive direction

The Lucas Group is a space-filling block of three complete Cell3 volumes along each primitive direction. It contains $3^3=27$ Cell3 volumes and has outer basis $B_G=3B$. This operation does not subdivide one cell. It aggregates complete cells while preserving orientation and shape.

```

B_G = 3B
Cell3 volumes = 3^3 = 27
Outer face area = 3^2 times one Cell3 face area
Group volume = 3^3 V0 = 27 V0

```

Measure	Cell3	Lucas Group	Factor
a outer edge	0.334179679059 nm	1.002539037177 nm	3
b outer edge	0.586410890411 nm	1.759232671233 nm	3
c outer edge	0.792319765043 nm	2.376959295128 nm	3
ab outer face	0.194869165460 nm ²	1.753822489144 nm ²	9
ac outer face	0.261106235790 nm ²	2.349956122112 nm ²	9
bc outer face	0.357809539129 nm ²	3.220285852157 nm ²	9
Volume	0.117914891912 nm ³	3.183702081617 nm ³	27

One Lucas Group: 27 complete Cell3 volumes, outer basis 3B

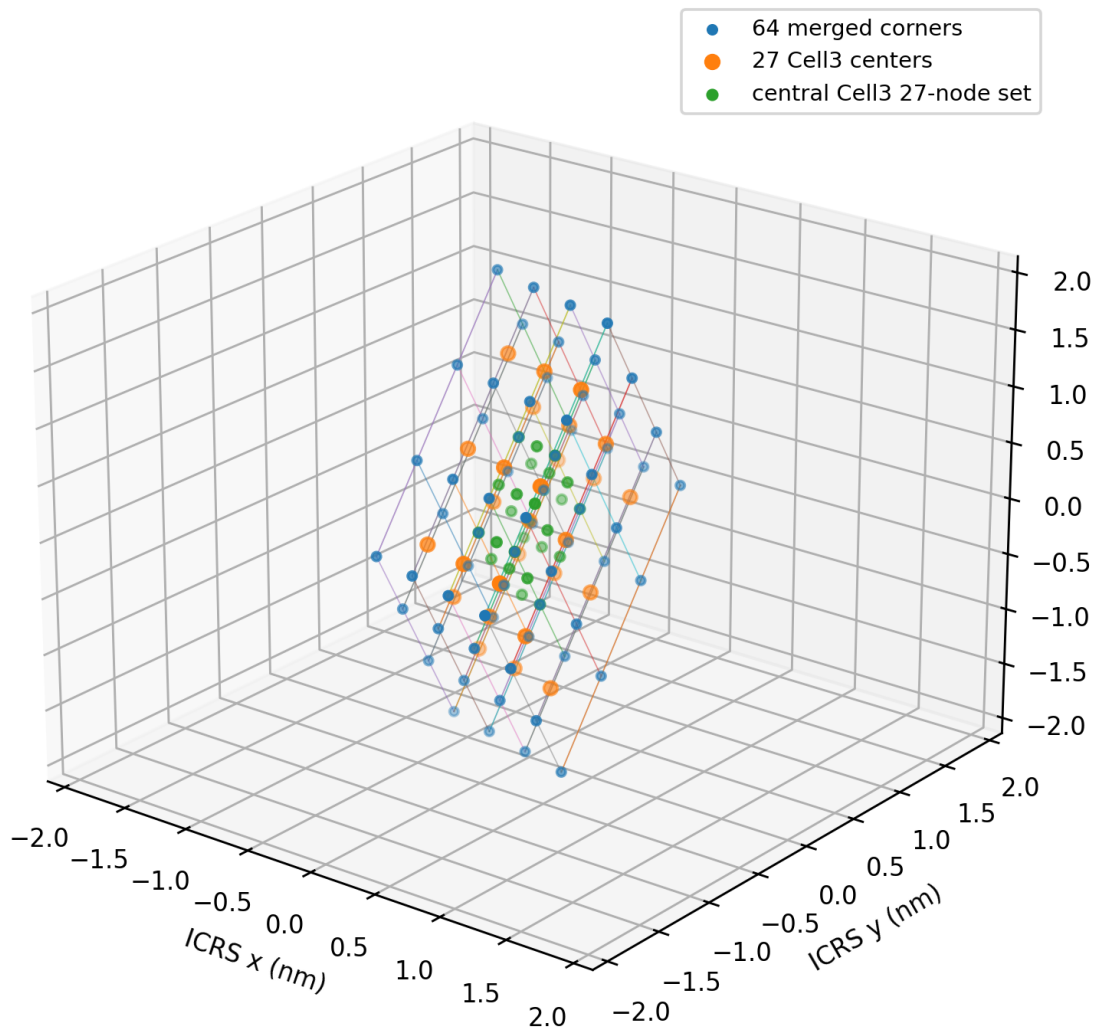


Figure 4.1. One complete Lucas Group. The exact space-filling grid contains 27 complete Cell3 volumes, 27 cell centers, 64 merged cell-corner nodes, and the central Cell3 27-node set.

4.2 The two different meanings of 27

The count 27 occurs twice but identifies different objects. A Lucas Group contains 27 complete volumes whose centers carry addresses g in $\{-1,0,1\}^3$. Separately, one Cell3 contains 27 midpoint nodes whose addresses L lie in the same integer cube. Equal combinatorial counts do not identify points with volumes. The distinction must remain explicit in every recursive calculation.

4.3 Collective phase sectors and internal connectivity

A periodic $3 \times 3 \times 3$ block has 27 translation sectors, one for each residue class in $\mathbb{Z}_3^3$. The cell-center adjacency graph has 27 vertices and 54 nearest-neighbor links. Its independent cycle count is $E - V + 1 = 28$. These cycles are the first extensive closed intercell paths available to the hierarchy.

```
m=3
vertices = m^3 = 27
edges = 3(m-1)m^2 = 54
cycle rank = edges - vertices + 1 = 28
```

5. The Second Aggregation and the Mass-Bridge Completion Theorem

5.1 Aggregation depth

Let g count repeated Lucas aggregation. The number of complete Cell3 volumes along each primitive axis is $m_g = 3^g$. The outer basis, area, and volume are therefore

$$\begin{aligned} m_g &= 3^g \\ B_g &= m_g \cdot B \\ A_g &= m_g^2 \cdot A_0 \\ V_g &= m_g^3 \cdot V_0 = 27^g \cdot V_0. \end{aligned}$$

At $g=2$, $m_2=9$. This block is a $3 \times 3 \times 3$ space-filling arrangement of complete Lucas Groups, or equivalently a $9 \times 9 \times 9$ arrangement of complete Cell3 volumes.

5.2 The 10-100-1000 node skeleton

A row of nine adjacent cells has ten boundary-node positions. A face formed from the Cartesian product of two such rows has $10^2=100$ boundary-grid nodes. The full three-dimensional corner lattice has $10^3=1000$ nodes.

$$\begin{aligned} \text{cells per axis} &= 3^2 = 9 \\ \text{corner positions per axis} &= 9 + 1 = 10 \\ \text{corner positions per face} &= 10^2 = 100 \\ \text{corner positions per volume} &= 10^3 = 1000 \end{aligned}$$

Mass-Bridge completion block: 9 x 9 x 9 Cell3 volumes

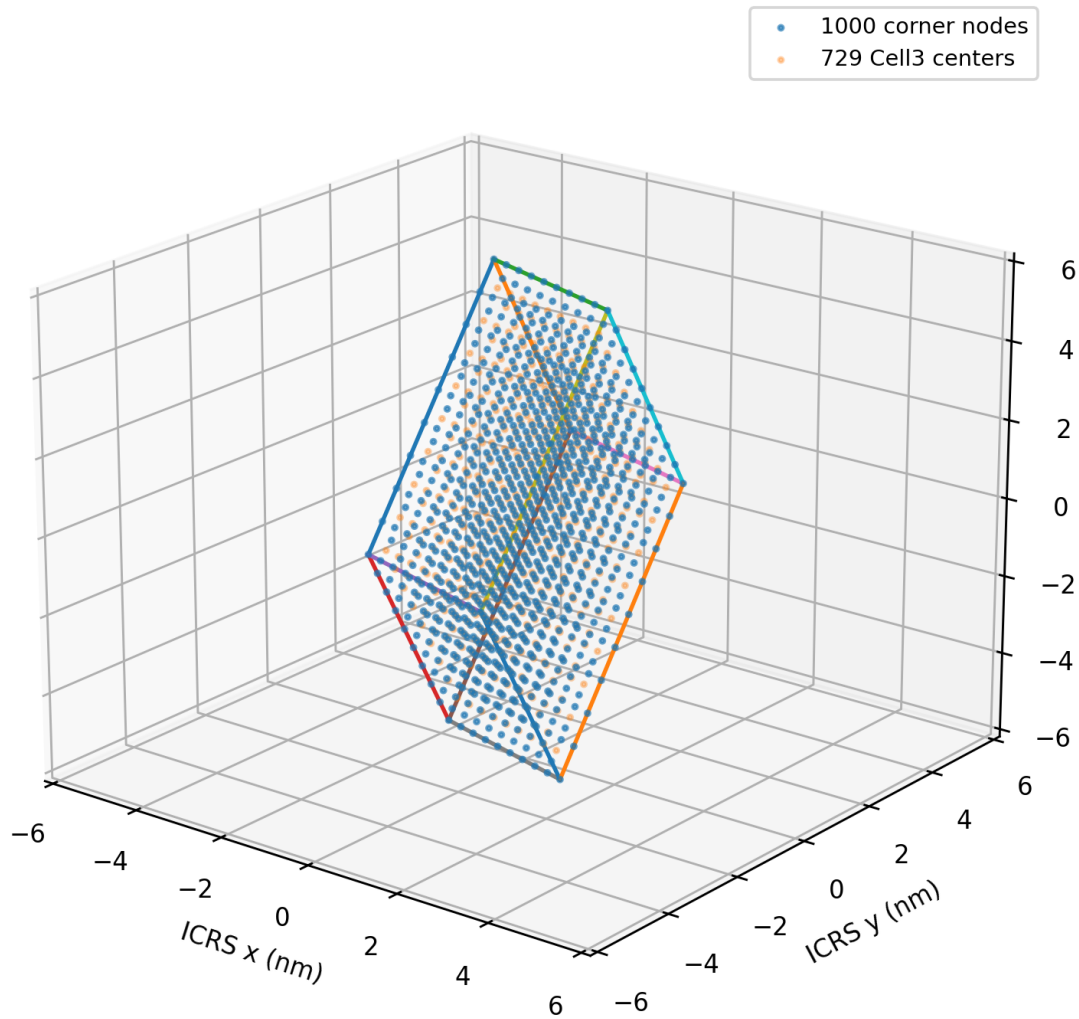


Figure 5.1. The $g=2$ completion block in ICRS coordinates. The exact unrefined corner lattice has ten positions along each primitive direction, one hundred per primitive face, and one thousand in the full block.

5.3 Uniqueness of the selected rung

The general group-rooted recursive grid has $3^g 2^n$ intervals per axis when g aggregation steps and n binary midpoint refinements are present. It therefore has $3^g 2^{n+1}$ node positions per axis. Setting this number equal to ten gives

$$\begin{aligned} 3^g 2^{n+1} &= 10 \\ 3^g 2^n &= 5 \\ (g, n) &= (2, 0). \end{aligned}$$

The exhaustive nonnegative-integer search returned $[(2, 0)]$. No other aggregation-refinement pair produces the required ten positions. The Mass Bridge therefore selects an unrefined $g=2$ block rather than a refined lower-level block.

5.4 Exact commutative closure

The geometric and dimensional multiplications close in parallel:

$$\begin{aligned} 10 \times 100 &= 1000 \\ (A_{ac}/A_{ab})(A_{bc}/A_{ac}) &= A_{bc}/A_{ab} \\ (13.399053419929)(137.035999177000) &= 1836.152673426000 \\ (\eta \alpha)(\alpha^{-1}) &= \eta \end{aligned}$$

The decimal factors are therefore represented internally as lattice cardinalities. Under this interpretation, the Mass Bridge does not merely constrain the local shape; it also names the first collective volume in which its complete dimensional composition is realized.

5.5 Full $g=2$ inventory

Object	Exact count or measure
Lucas Groups	27
Cell3 volumes	729
Cell3 volumes per axis	9
Unrefined corner nodes	1,000
Half-step nodes per axis	19
Merged half-step nodes	6,859
Constant-lambda sheets	55
Central $\lambda=0$ sheet nodes	271
Antipodal half-step node pairs	3,429
Cell-center translation sectors	729
Internal Cell3 interfaces	1,944
Exposed primitive faces	486
Internal/exposed interface ratio	4
Cell-center graph cycle rank	1,216
Strictly interior Cell3 volumes	343
Strictly interior half-step nodes	4,913

$g=2$ merged half-step lattice: 6,859 nodes on 55 lambda sheets

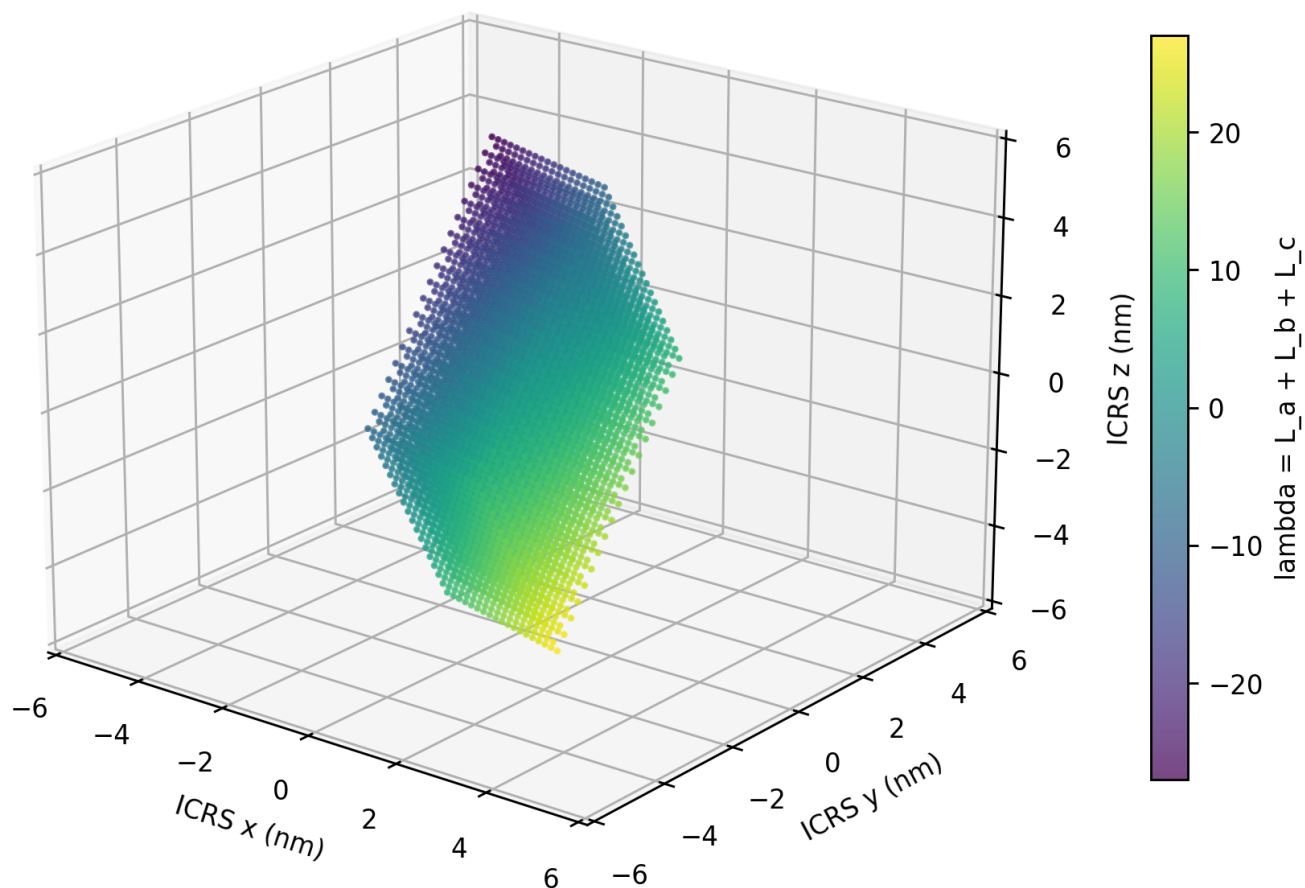


Figure 5.2. Complete $g=2$ half-step lattice. The 6,859 nodes occupy 55 constant-lambda sheets. Color encodes the exact integer lambda address.

6. Accumulated Base Dimensions

6.1 Direct scaling

Because every aggregation copies complete congruent Cell3 volumes, every length grows by 3^g , every area by 9^g , and every volume by 27^g . The shape ratios and included angles remain unchanged.

g	Object	Cell3 count	Outer a,b,c (nm)	Volume (nm ³)	Body diagonal (nm)
0	Cell3	1	0.334180, 0.586411, 0.792320	0.117914892	1.343654
1	Lucas Group	27	1.002539, 1.759233, 2.376959	3.18370208	4.030961
2	Mass-Bridge block	729	3.007617, 5.277698, 7.130878	85.9599562	12.092883
3	Bulk block	19,683	9.022851, 15.833094, 21.392634	2320.91882	36.278648
4	Extended bulk	531,441	27.068554, 47.499282, 64.177901	62664.8081	108.835943
5	Larger bulk	14,348,907	81.205662, 142.497846, 192.533703	1691949.82	326.507830

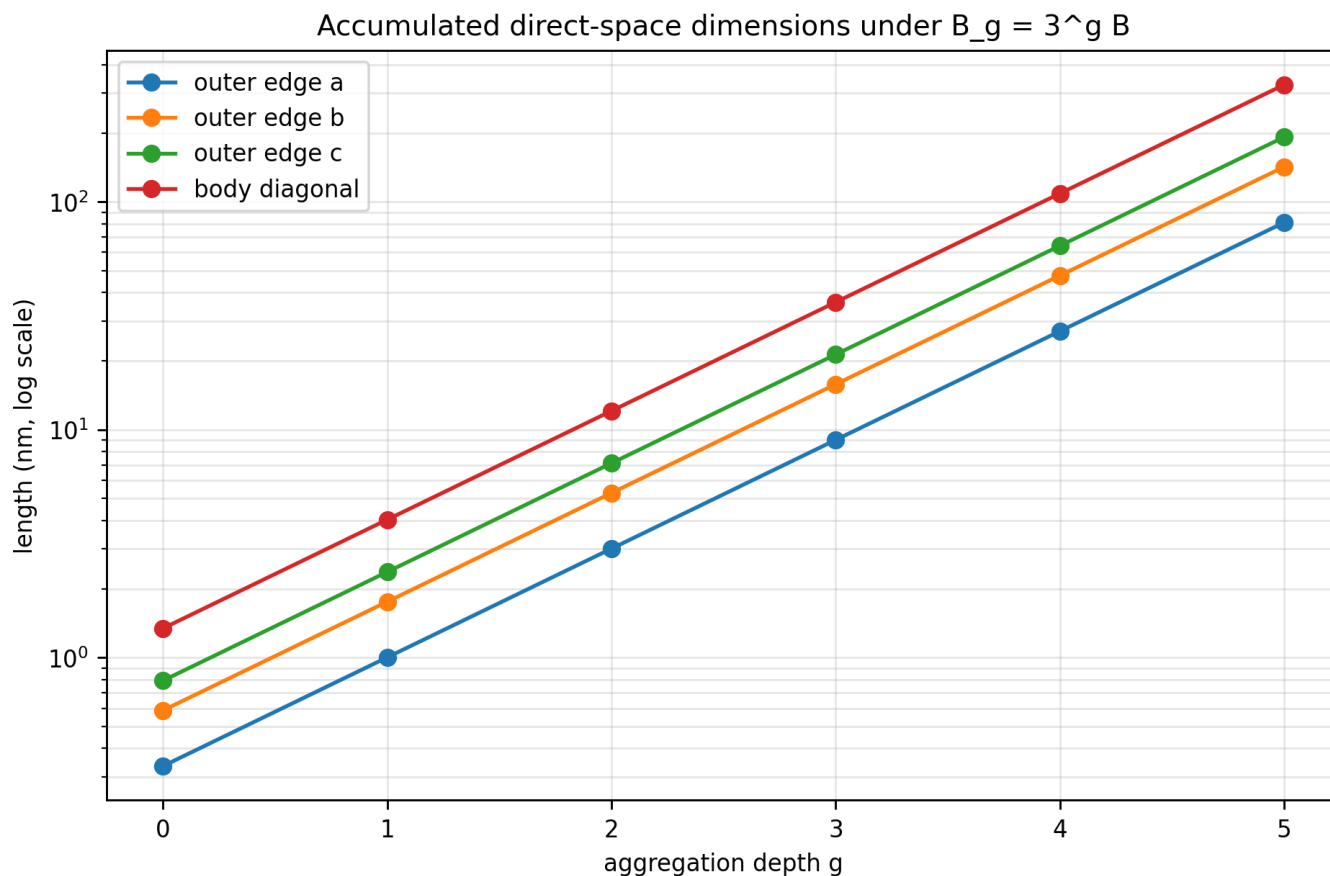


Figure 6.1. Accumulated direct-space dimensions. All plotted values are exact multiples of the locked Cell3 edge lengths and body diagonal.

6.2 The complete $g=2$ dimensions

Measure	Value
Outer edges a,b,c	3.007617111532 nm, 5.277698013700 nm, 7.130877885384 nm
Outer face areas ab,ac,bc	15.784402402299 nm ² , 21.149605099007 nm ² , 28.982572669413 nm ²
Volume	85.959956203662 nm ³
Opposite-face spans a,b,c	2.965918767259 nm, 4.064376417492 nm, 5.445879673667 nm
Body diagonal	12.092882595083 nm

These dimensions describe the adopted nanometre realization. If the uniform scale is changed, every direct dimension changes by its dimensional degree while the $g=2$ count theorem and all projective ratios remain exact.

7. General Volume-Filling Hierarchy

7.1 Closed-form count laws

Set $m=3^g$. The unrefined and half-step lattices then have exact polynomial counts:

```

Cell3 volumes:  $N_{\text{cell}} = m^3$ 
Unrefined corner nodes:  $N_{\text{corner}} = (m+1)^3$ 
Half-step nodes:  $Q_g = (2m+1)^3$ 
Lambda sheets:  $S_g = 6m+1$ 
Central sheet nodes:  $C_g = 3m^2+3m+1$ 
Antipodal pairs:  $P_g = (Q_g-1)/2$ 

```

The central-sheet formula counts integer triples in $[-m,m]^3$ whose components sum to zero. The antipodal-pair formula removes the unique body center and pairs every remaining node with its negative address.

7.2 Interfaces, exposed faces, and graph cycles

Internal interfaces: $I_g = 3(m-1)m^2$
 Exposed primitive faces: $F_g = 6m^2$
 $I_g/F_g = (m-1)/2$
 Cell-center cycle rank: $\beta_1 = I_g - m^3 + 1$

The interface ratio grows linearly with the number of cells per axis. At $g=3$, $m=27$ and $I_g/F_g=13$, exactly matching the 13 local antipodal node channels of one Cell3. This equality is a structural recurrence between local channel count and collective interface dominance.

7.3 Complete hierarchy table

g	m	Cell3	Corner nodes	Half-step nodes	Sheets	Central sheet	Pairs	Interior Cell3	I/F
0	1	1	8	27	7	7	13	0.000%	0.000
1	3	27	64	343	19	37	171	3.704%	1.000
2	9	729	1,000	6,859	55	271	3,429	47.051%	4.000
3	27	19,683	21,952	166,375	163	2,269	83,187	79.383%	13.000
4	81	531,441	551,368	4,330,747	487	19,927	2,165,373	92.774%	40.000
5	243	14,348,907	14,526,784	115,501,303	1,459	177,877	57,750,651	97.551%	121.000

Growth of volumes, nodes, and laminar sheets

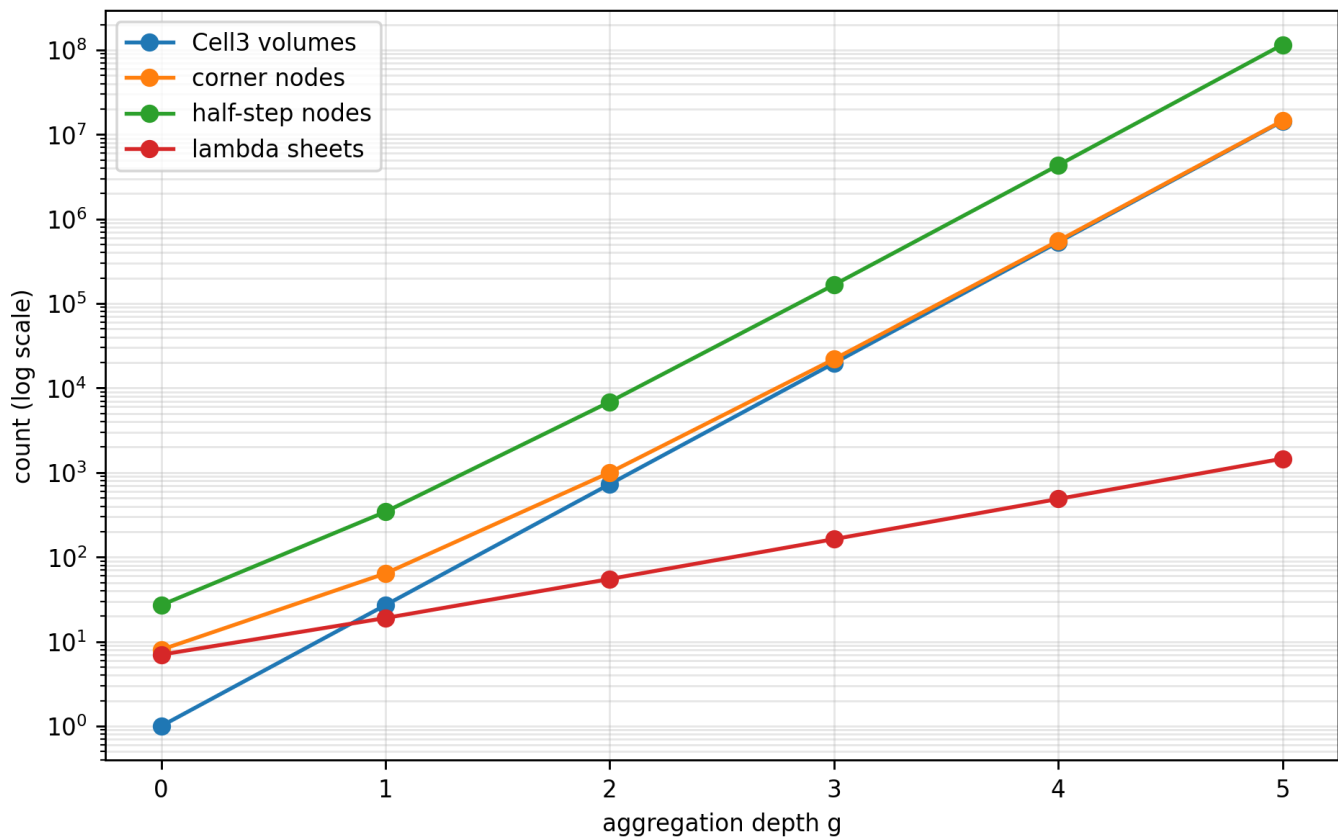


Figure 7.1. Count growth across aggregation depth. Cell volumes grow as 27^g , while node and sheet counts follow their exact polynomial formulas in $m=3^g$.

7.4 Exact sheet-to-volume recursion

Because $S_g = 6 \cdot 3^g + 1$ and the next half-step lattice has $2 \cdot 3^{(g+1)+1}$ positions per axis, the two counts are identical:

$S_g = 6 \cdot 3^g + 1 = 2 \cdot 3^{(g+1)+1} = Q_{-}(g+1)^{(1/3)}$
 therefore $S_{g^3} = Q_{-}(g+1)$.

The complete laminar count of one hierarchy level becomes the one-axis node count of the next, and its cube becomes the full next-level half-step lattice. This is not an approximate scaling rule; it is an exact recursion.

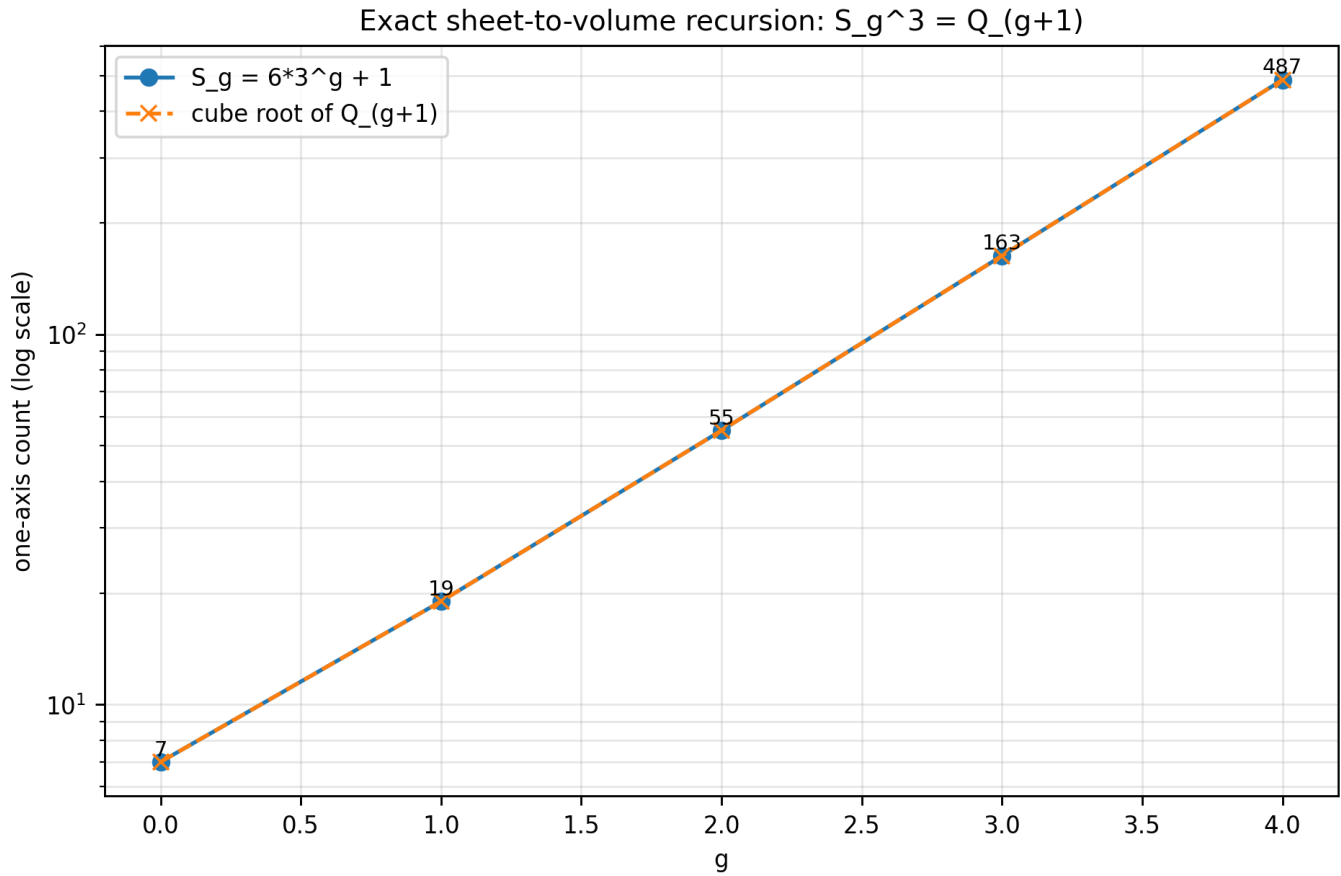


Figure 7.2. Numerical verification of the exact identity $S_g^3 = Q_{(g+1)}$. The two plotted sequences coincide at every displayed level.

7.5 Parity conservation and residual one

Every half-step lattice has an odd number of positions per axis. Under polarity $(-1)^{(L_a+L_b+L_c)}$, the total signed population is always -1:

$$\sum_{L_a=-m}^m \sum_{L_b=-m}^m \sum_{L_c=-m}^m (-1)^{(L_a+L_b+L_c)} = -1, \text{ for odd } m=3^g.$$

As the lattice grows, positive and negative populations become nearly equal, but their absolute difference remains exactly one. This supplies a scale-independent residual that survives arbitrary aggregation.

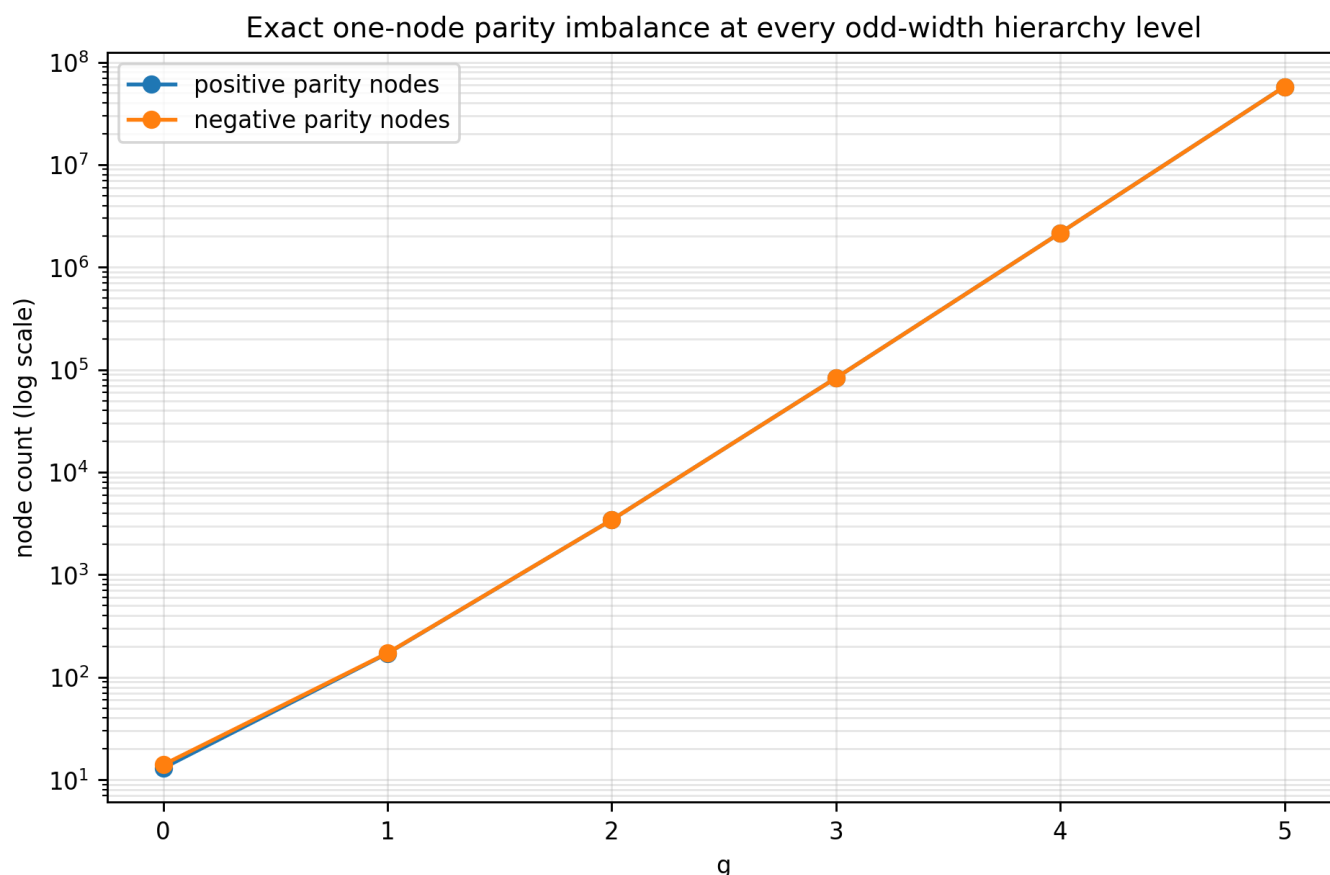


Figure 7.3. Positive and negative parity populations. Both grow extensively, while their exact difference remains one node at every level.

8. From Local Object to Collective Field Volume

8.1 $g=0$: local geometric object

A single Cell3 carries the complete local shape: three edge vectors, three face channels, six boundary planes, a 27-node midpoint grammar, seven lambda sheets, thirteen antipodal pairs, and the direct-reciprocal metric pair. It is geometrically complete but has no completely surrounded constituent Cell3 and no extensive intercell network.

8.2 $g=1$: first collective resonator

The Lucas Group is the first aggregation with a central Cell3 surrounded on all six primitive sides. It supports 27 cell-center translation sectors, 54 internal interfaces, 28 independent adjacency cycles, a 343-node half-step lattice, and 19 lambda sheets. The local 27-node grammar becomes an internal sample block of a larger exact lattice.

8.3 $g=2$: Mass-Bridge-complete finite object

At $g=2$, the hierarchy first realizes the Mass-Bridge cardinalities. The lattice is already node-bulk-dominated: 4,913 of 6,859 half-step nodes are strictly internal, or 71.63 percent. The Cell3 volume fraction is $343/729=47.05$ percent, so the block lies at a transition: its node field is bulk-dominated while its complete-cell population is still nearly balanced between interior and boundary layers.

8.4 $g=3$: bulk-dominated field volume

At $g=3$, 15,625 of 19,683 Cell3 volumes are strictly interior, or 79.38 percent, and 148,877 of 166,375 half-step nodes are interior, or 89.48 percent. The internal-to-exposed interface ratio reaches 13. The hierarchy has crossed into a regime in which the bulk, rather than the boundary, controls most of the available local environments and intercell pathways.

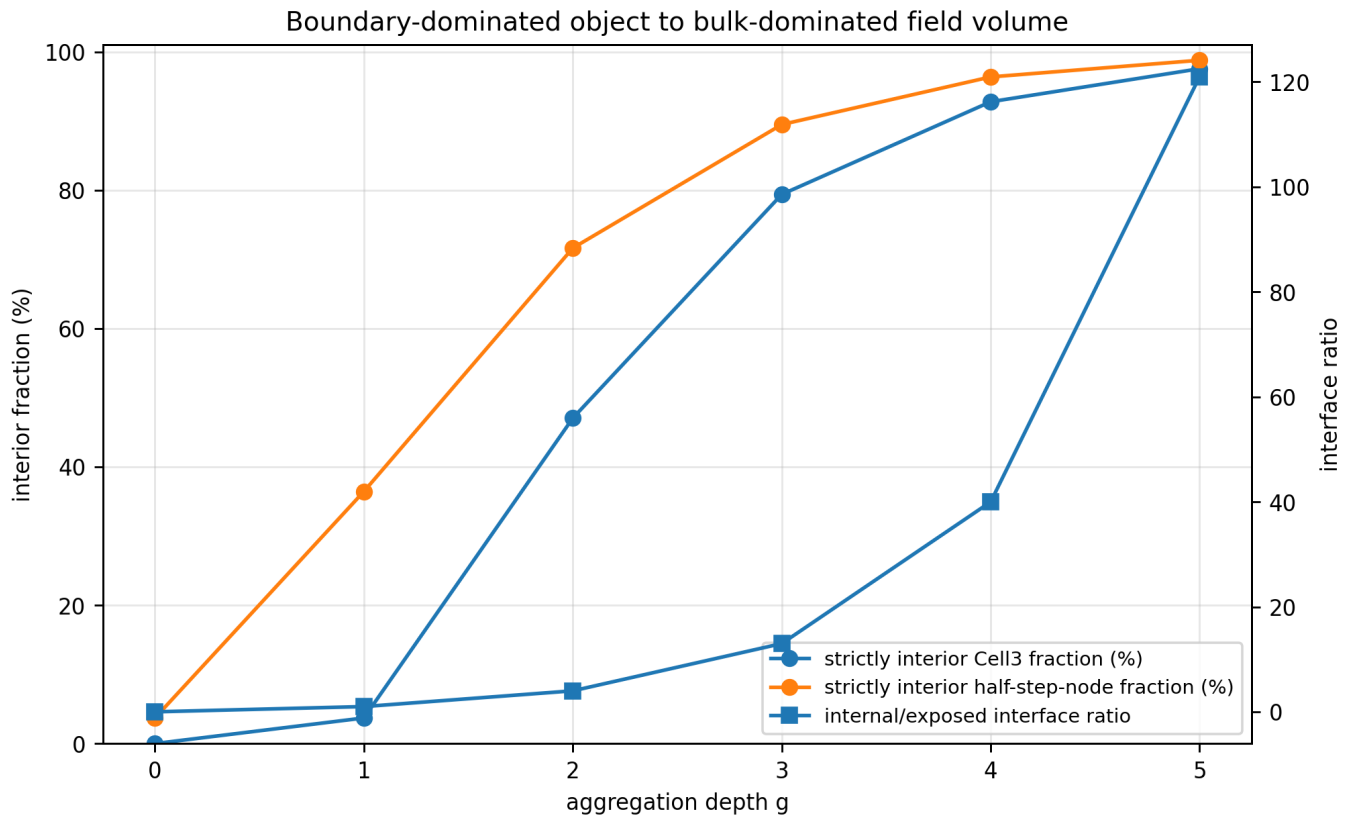


Figure 8.1. Quantitative object-to-field transition. The node population becomes bulk-dominated at $g=2$; the complete-cell population becomes strongly bulk-dominated at $g=3$.

8.5 Interpretation of increasing complexity

The hierarchy adds complexity without altering the primitive shape. New behavior comes from relational multiplicity: more internal interfaces, more closed graph cycles, more phase sectors, more laminar sheets, and a denser reciprocal mode set. A local metric becomes a collective translation problem; a finite translation problem becomes a bulk field volume. The progression therefore supplies a concrete mechanism by which increasingly complex physical objects and fields can be represented while preserving one locked geometric kernel.

9. Reciprocal Space and Harmonic Families

9.1 Allowed reciprocal vectors

For a periodic block with outer basis mB , an integer mode index q gives the reciprocal vector

$$k_q(g) = 2 \pi i (mB)^{(-T)} q = (2 \pi i / 3^g) B^{(-T)} q.$$

Its squared magnitude is evaluated entirely by the locked reciprocal metric:

$$|k_q(g)|^2 = (2 \pi i / 3^g)^2 q^T R q.$$

The integer quadratic form $q^T R q$ determines the anisotropic ordering of modes. Aggregation changes only the common scale factor 3^{-g} ; it does not change the ordering or directional structure.

9.2 Direct and reciprocal scaling table

Quantity	One aggregation	One dyadic refinement
Direct length	x 3	/ 2
Area	x 9	/ 4
Volume	x 27	/ 8
Reciprocal wavenumber	/ 3	x 2
Quadratic eigenvalue	/ 9	x 4
Collective sector count	x 27	x 8 local subcells

Linear-dispersion family: $\omega(g,n)/\omega_0 = 2^n/3^g$

Quadratic family: $E(g,n)/E_0 = 4^n/9^g$.

The factors of three and two describe different operations. Ternary aggregation creates larger collective objects and spectral subharmonics. Binary subdivision creates finer internal structure and spectral harmonics. Mixing them without preserving their roles would erase the hierarchy grammar.

9.3 Enumerated low-order reciprocal spectrum

Rank	q representatives	$q^T R q \text{ (nm}^{-2}\text{)}$	Deg.	$ k \text{ g=0}$	$ k \text{ g=1}$	$ k \text{ g=2}$
1	(0, 0, -1); (0, 0, 1)	2.731171275	2	10.383753	3.461251	1.153750
2	(0, -1, -1); (0, 1, 1)	3.004218418	2	10.890445	3.630148	1.210049
3	(0, -1, 0); (0, 1, 0)	4.903398312	2	13.913246	4.637749	1.545916
4	(0, -1, -2); (0, 1, 2)	6.567381074	2	16.101857	5.367286	1.789095
5	(-1, 0, 0); (1, 0, 0)	9.208025529	2	19.066155	6.355385	2.118462
6	(-1, 0, -1); (1, 0, 1)	10.648820733	2	20.503624	6.834541	2.278180
7	(0, 0, -2); (0, 0, 2)	10.924685099	2	20.767505	6.922502	2.307501
8	(-1, -1, -1); (1, 1, 1)	10.924937481	2	20.767745	6.922582	2.307527
9	(0, -2, -2); (0, 2, 2)	12.016873673	2	21.780890	7.260297	2.420099
10	(0, -1, 1); (0, 1, -1)	12.264920756	2	22.004537	7.334846	2.444949
11	(0, -2, -1); (0, 2, 1)	13.084062187	2	22.727474	7.575825	2.525275
12	(-1, -1, -2); (1, 1, 2)	13.197724065	2	22.825978	7.608659	2.536220
13	(-1, 0, 1); (1, 0, -1)	13.229572876	2	22.853503	7.617834	2.539278
14	(-1, 1, 1); (1, -1, -1)	13.499550414	2	23.085513	7.695171	2.565057
15	(-1, 1, 0); (1, -1, 0)	14.108354237	2	23.600328	7.866776	2.622259
16	(-1, -1, 0); (1, 1, 0)	14.114493447	2	23.605463	7.868488	2.622829

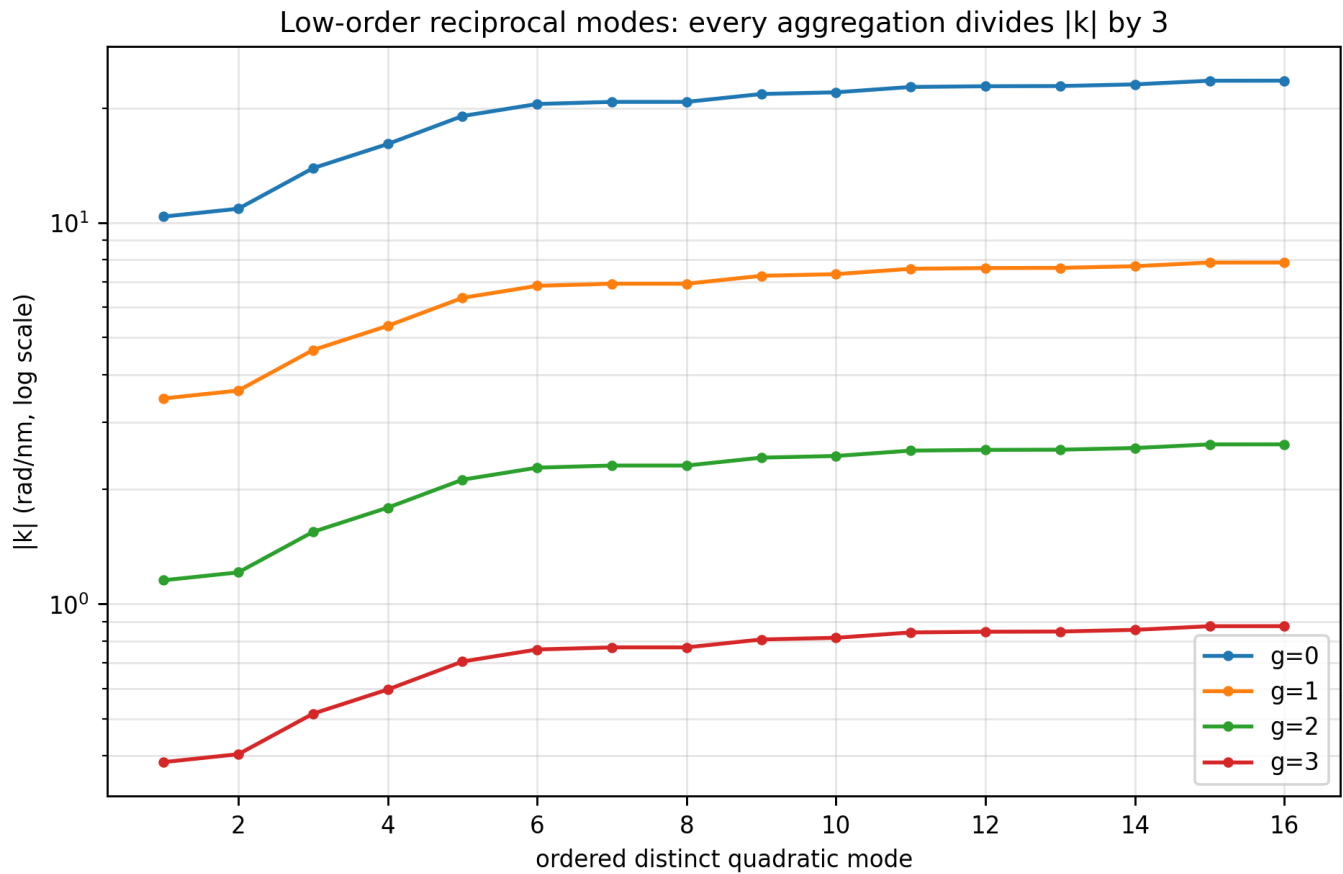


Figure 9.1. First sixteen distinct reciprocal magnitudes. The complete anisotropic ordering is set by $q^T R q$, and each aggregation level divides every magnitude by three.

9.4 Collective phase sectors

A block with m cells per axis has m^3 translation sectors. At $g=1$ there are 27 sectors; at $g=2$ there are 729; at $g=3$ there are 19,683. Each sector is a discrete phase character over the cell-center translation group. The residue of an integer reciprocal index modulo m labels the collective cell-to-cell phase arrangement.

10. Harmonics and Subharmonics of Established Physics

10.1 Controlled comparison method

The comparison dictionary was declared before examining the residual ranking. It contains two classes: integer powers $2^n 3^g$ over a bounded exponent range, and canonical hierarchy counts derived from the exact formulas in this paper, including their explicit one-, two-, and three-dimensional decimal lifts. Every measured quantity is compared in logarithmic space, so upward and downward relative deviations are treated symmetrically. The two Mass-Bridge anchors are retained as construction checks and excluded from independent discovery significance.

A close numerical residual is not automatically a physical law. A candidate gains weight when the same hierarchy exponents recur across independent observables, when the mapping follows dimensional roles fixed before the comparison, and when residuals organize rather than merely shrink.

10.2 Strongest retained candidates

Measured quantity	Observed	Hierarchy candidate	Relative residual	Abs. ppm
electron to shielded helion magnetic-moment ratio	864.058255000	$2^5 \cdot 3^3 = 864.000000000$	-0.006742%	67.4
triton/electron mass ratio	5496.921535510	$100 \cdot 55$ (g=2 sheet count) = 5500.000000000	+0.056003%	560.0
alpha/electron mass ratio	7294.299541710	$10 \cdot 729$ (g=2 Cell3 count) = 7290.000000000	-0.058944%	589.4
helion/electron mass ratio	5495.885279840	$100 \cdot 55$ (g=2 sheet count) = 5500.000000000	+0.074869%	748.7
charged-pion/electron mass ratio	273.132439532	$3^7 / 2^3 = 273.375000000$	+0.088807%	888.1
charged-pion/electron mass ratio	273.132439532	g=2 central sheet = 271 = 271.000000000	-0.780735%	7807.3
tau/electron mass ratio	3477.230000000	g=2 antipodal pairs = 3429 = 3429.000000000	-1.387024%	13870.2

Selected hierarchy and harmonic comparisons with established dimensionless quantities

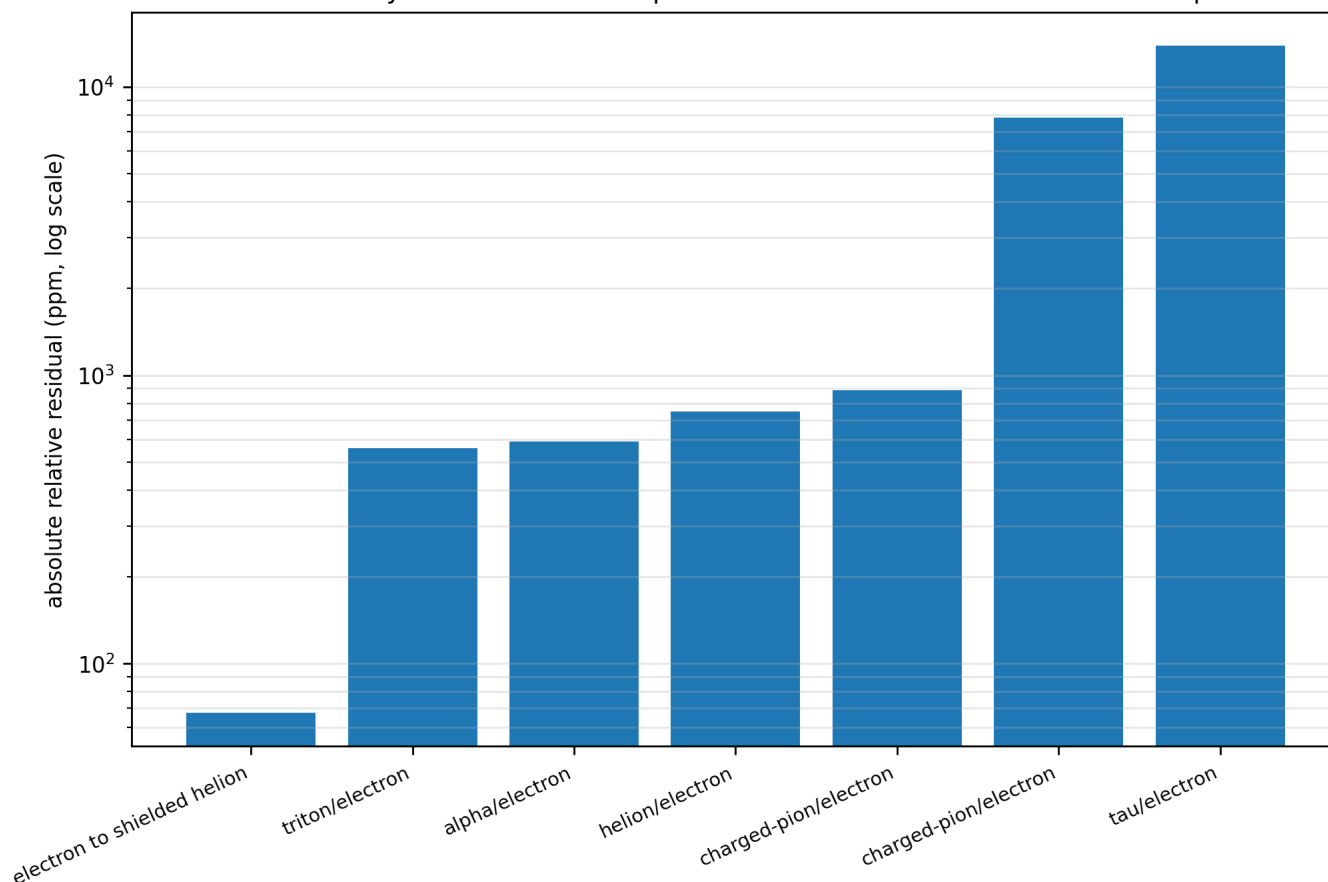


Figure 10.1. Absolute residuals for the selected hierarchy comparisons. The 864.058255 to 864 relation is the strongest non-anchor candidate in this set.

10.3 Statistical reading of the 864 candidate

The electron-to-shielded-helion magnetic-moment ratio is 864.058255, while $2^5 \cdot 3^3 = 864$. The relative residual is -6.7420×10^{-5} , or -67.4 parts per million. Under the local log-uniform approximation used for the initial scan, the nominal probability is approximately 0.0067. Correcting across the 26 tested non-anchor quantities gives a false-discovery value near 0.175. The relation is therefore the leading retained candidate, but it is not yet an independently established quantization law.

10.4 Nuclear-count candidates at the Mass-Bridge level

Two candidates specifically reuse $g=2$ structural counts. The 55-sheet count, multiplied by the two-dimensional factor 100, gives 5500 and lies within 560 ppm of the triton/electron mass ratio and 749 ppm of the helion/electron mass ratio. The 729 Cell3 volume count, multiplied by the one-dimensional factor 10, gives 7290 and lies within 589 ppm of the alpha/electron mass ratio. These repeated uses of the same $g=2$ object are more informative than unrelated nearest integers, because the count source and dimensional multiplier are fixed by the same completion block.

10.5 Pion and tau structural counts

The charged-pion/electron mass ratio computed from 139.57039 MeV and the 2022 CODATA electron mass energy is 273.132439532. The dyadic-ternary candidate $3^{7/2}/3=273.375$ differs by +0.088807 percent. The $g=2$ central-sheet count 271 differs by -0.780735 percent. The tau/electron mass ratio 3477.23 differs from the $g=2$ antipodal-pair count 3429 by -1.387024 percent. These are structurally interpretable residuals, but their magnitude is larger than the leading magnetic-moment and nuclear-mass candidates.

10.6 Why unrestricted matching is rejected

The combined powers of two and three become dense on a logarithmic axis when the exponent range is allowed to grow without bound. Canonical hierarchy counts and decimal lifts add further density. A search that simply reports the nearest available state will eventually find an apparently close value for almost any observation. The correct program is derive, compare, inspect residuals, identify repeated exponents or count families, and then test those families on new observables without changing the map.

11. Testable Consequences and Research Program

11.1 Minimal reciprocal-spectrum test

The next decisive calculation is the complete low-order list of $q^T R q$ values for $g=1$, $g=2$, and $g=3$, including degeneracies and phase-sector residues. One global scale may be assigned to compare dimensioned spectra. No nearest-state search, particle-specific coefficient, or hierarchy-dependent rescaling is permitted. A valid correspondence must preserve ordering, degeneracy structure, and residual organization across multiple independent spectral families.

11.2 Boundary-to-bulk transition tests

The hierarchy predicts distinct structural thresholds. At $g=1$ the first completely surrounded Cell3 appears. At $g=2$ the Mass-Bridge cardinality closure appears and the half-step node population becomes bulk-dominated. At $g=3$ the complete Cell3 population becomes strongly bulk-dominated and the interface ratio reaches 13. Physical models built on the hierarchy should therefore show qualitative changes at these exact levels rather than at arbitrary cell counts.

11.3 Sheet inheritance

The identity $S_g = Q_{g+1}$ predicts that the complete laminar count of one level is the exact one-axis address count of the next. A field construction using sheet phases should be testable for this inheritance: level- g sheet labels should map directly into level- $(g+1)$ node coordinates without interpolation or fitted phase offsets.

11.4 Residual recursion

For every retained physics candidate, the residual should be carried forward as data rather than discarded. The first question is whether residuals align with a secondary hierarchy count, a reciprocal quadratic mode, a parity sector, or a boundary-interface correction. The second is whether the same correction law predicts residuals for other quantities. A correction that is introduced for only one object is not accepted.

12. Principal Findings

Finding	Status	Result
Mass Bridge fixes Cell3 shape	Locked source	Two independent face ratios and one exact closure ratio; uniform scale only remains
Lucas aggregation	Exact derivation	$B_g=3^g B$ and 27^g complete Cell3 volumes
Unique selected rung	Exact theorem	$3^g 2^{n+1}=10$ has only $(g,n)=(2,0)$
Dimensional closure	Exact theorem	10, 100, and 1000 are $g=2$ line, face, and volume corner-node counts
Mass-Bridge completion object	Exact derivation	9B block with 27 Lucas Groups, 729 Cell3 volumes, 1,000 corner nodes, 6,859 half-step nodes, 55 sheets
Sheet-to-volume recursion	Exact theorem	$S_g=Q_{g+1}$
Parity residual	Exact theorem	Signed node population equals -1 at every aggregation level
Object-to-field transition	Structural interpretation	$g=2$ node-bulk transition; $g=3$ complete-cell bulk transition
Reciprocal hierarchy	Exact derivation	$ k $ scales as 3^g and dyadic refinement as 2^n
864 candidate	Empirical comparison	-67.4 ppm residual; retained after correction as a candidate, not a completed law

The dominant conclusion is that the Mass Bridge and recursive volume filling are not separate constructions. The locked face ratios identify the local shape, while their 10-100-1000 projective multipliers identify the unique collective volume in which the dimensional composition is represented by exact lattice cardinalities.

13. Conclusions

Establishing the Mass Bridge as a shape-forcing constraint changes the direction of analysis. Cell3 is treated as a locked geometric primitive. Space-filling copies then generate a deterministic hierarchy whose new features arise from accumulated relationships rather than altered local geometry.

The first Lucas Group creates collective translation sectors and closed intercell cycles. The second aggregation creates the exact Mass-Bridge completion object. The third aggregation creates a strongly bulk-dominated field volume and an interface ratio equal to the thirteen local antipodal channels. At every level the direct and reciprocal descriptions remain linked by the same basis.

The $g=2$ object is the principal new result: a 9B block containing 27 Lucas Groups and 729 Cell3 volumes, with exactly 10, 100, and 1000 unrefined corner-node cardinalities in one, two, and three dimensions. Those counts multiply the locked face ratios into ϵ , ϵ^{-1} , and ϵ . The construction therefore joins local projective physics with collective volume geometry in one exact hierarchy.

The broader physics program now has a deterministic target. Enumerate the reciprocal quadratic forms, preserve degeneracies and phase sectors, apply one global dimensional scale, and test measured spectra without nearest-state searches or object-specific coefficients. Residuals are then analyzed recursively for secondary hierarchy structure. This program converts the volume-filling hypothesis into a sequence of explicit calculations and falsifiable comparisons.

References and Source Record

1. Krytronx. K-Field Mass-Bridge Locked Projective Ratio System. JSON release epoch 1785176592. Source SHA-256: a5ba3ddf21a20e4528c19609a3061fceca60f53f76224f101c9779c24809d7ea.

2. Krytronx. K-Field Fractional Recursive Quanta - Lucas Group. Machine-readable release epoch 1784239728. Source SHA-256: 75631c25c2ccc16ca116b01a7a629f4c22d3caac34b4be43ccd16845ecef412e.

3. Tiesinga, E.; Mohr, P. J.; Newell, D. B.; Taylor, B. N. The 2022 CODATA Recommended Values of the Fundamental Physical Constants, Web Version 9.0, National Institute of Standards and Technology, 2024.

4. SciPy 1.17.0 physical_constants table, using the 2022 CODATA recommended values.

5. Particle Data Group. Review of Particle Physics, 2025 update. Charged-pion mass value used for the pion/electron ratio comparison.

Appendix A. Complete Locked Basis, Metrics, and Planes

Basis component	a	b	c
ICRS x	0.327501242055891	0.035330775843828	-0.010129317626729
ICRS y	-0.054897007884006	0.207319850440192	-0.389787702366260
ICRS z	-0.037487502894730	-0.547401085388508	-0.689734408367268

Direct metric row	Column a	Column b	Column c
0	0.111676057896050	0.020710333280606	0.043937235096220
1	0.020710333280606	0.343877732392787	0.296392758961523
2	0.043937235096220	0.296392758961523	0.627770610077296

Reciprocal metric row	Column a*	Column b*	Column c*
0	9.208025529457549	0.001534802428193	-0.645188035699939
1	0.001534802428193	4.903398312367987	-2.315175584416389
2	-0.645188035699939	-2.315175584416389	2.731171274711695

Channel	Coefficient	Primitive coordinate	Normal x,y,z	Area	Spacing	Center-to-face
a	-1	-0.5	-0.995965190309, 0.083602162507, -0.032619290564	0.357809539129	0.329546529695	0.164773264848
a	1	0.5	-0.995965190309, 0.083602162507, -0.032619290564	0.357809539129	0.329546529695	0.164773264848
b	-1	-0.5	-0.089052594102, -0.866576769291, 0.491033948326	0.261106235790	0.451597379721	0.225798689861
b	1	0.5	-0.089052594102, -0.866576769291, 0.491033948326	0.261106235790	0.451597379721	0.225798689861
c	-1	-0.5	0.194092200808, 0.913177168821, 0.358379234791	0.194869165460	0.605097741519	0.302548870759
c	1	0.5	0.194092200808, 0.913177168821, 0.358379234791	0.194869165460	0.605097741519	0.302548870759

Appendix B. Complete 27-Node Ledger

Address	lambda	Polarity	Class	ICRS position (nm)
(-1, -1, -1)	-3	-1	corner	-0.176351350136495, 0.118682429905037, 0.637311498325253
(-1, -1, 0)	-2	1	edge center	-0.181416008949859, -0.076211421278093, 0.292444294141619
(-1, -1, 1)	-1	-1	corner	-0.186480667763224, -0.271105272461223, -0.052422910042015
(-1, 0, -1)	-2	1	edge center	-0.158685962214581, 0.222342355125133, 0.363610955630999
(-1, 0, 0)	-1	-1	face center	-0.163750621027945, 0.027448503942003, 0.018743751447365
(-1, 0, 1)	0	1	edge center	-0.168815279841310, -0.167445347241127, -0.326123452736269
(-1, 1, -1)	-1	-1	corner	-0.141020574292667, 0.326002280345229, 0.089910412936745
(-1, 1, 0)	0	1	edge center	-0.146085233106031, 0.131108429162099, -0.254956791246889
(-1, 1, 1)	1	-1	corner	-0.151149891919396, -0.063785422021031, -0.599823995430523
(0, -1, -1)	-2	1	edge center	-0.012600729108550, 0.091233925963034, 0.618567746877888
(0, -1, 0)	-1	-1	face center	-0.017665387921914, -0.103659925220096, 0.273700542694254
(0, -1, 1)	0	1	edge center	-0.022730046735279, -0.298553776403226, -0.071166661489380
(0, 0, -1)	-1	-1	face center	0.005064658813364, 0.194893851183130, 0.344867204183634
(0, 0, 0)	0	1	body center	0.000000000000000, 0.000000000000000, 0.000000000000000
(0, 0, 1)	1	-1	face center	-0.005064658813364, -0.194893851183130, -0.344867204183634
(0, 1, -1)	0	1	edge center	0.022730046735279, 0.298553776403226, 0.071166661489380
(0, 1, 0)	1	-1	face center	0.017665387921914, 0.103659925220096, -0.273700542694254
(0, 1, 1)	2	1	edge center	0.012600729108550, -0.091233925963034, -0.618567746877888
(1, -1, -1)	-1	-1	corner	0.151149891919396, 0.063785422021031, 0.599823995430523
(1, -1, 0)	0	1	edge center	0.146085233106031, -0.131108429162099, 0.254956791246889
(1, -1, 1)	1	-1	corner	0.141020574292667, -0.326002280345229, -0.089910412936745
(1, 0, -1)	0	1	edge center	0.168815279841310, 0.167445347241127, 0.326123452736269
(1, 0, 0)	1	-1	face center	0.163750621027945, -0.027448503942003, -0.018743751447365
(1, 0, 1)	2	1	edge center	0.158685962214581, -0.222342355125133, -0.363610955630999
(1, 1, -1)	1	-1	corner	0.186480667763224, 0.271105272461223, 0.052422910042015
(1, 1, 0)	2	1	edge center	0.181416008949859, 0.076211421278093, -0.292444294141619
(1, 1, 1)	3	-1	corner	0.176351350136495, -0.118682429905037, -0.637311498325253

Appendix C. Hierarchy Ledger through g=6

g	m	Cells	Corners	Half nodes	Sheet s	Center sheet	Pairs	Interfaces	Exposed	Cycle rank	Interior cells %	Interior nodes %
0	1	1	8	27	7	7	13	0	6	0	0.000000	3.703704
1	3	27	64	343	19	37	171	54	54	28	3.703704	36.443149
2	9	729	1000	6859	55	271	3429	1944	486	1216	47.050754	71.628517
3	27	19683	21952	166375	163	2269	83187	56862	4374	37180	79.383224	89.482795
4	81	531441	551368	4330747	487	19927	2165373	1574640	39366	1043200	92.773986	96.363999
5	243	14348907	14526784	115501303	1459	177877	57750651	42869574	354294	28520668	97.551131	98.773020
6	729	387420489	389017000	3105745579	4375	1596511	1552872789	1160667144	3188646	773246656	99.179211	99.589323

Appendix D. Low-Order Reciprocal Quadratic Spectrum

Ran k	q vectors	Degene racy	q ^{^T} R q	sqrt	2pi sqrt	g=1 k	g=2 k	g=3 k
1	(0, 0, -1); (0, 0, 1)	2	2.7311712747 12	1.652625570 028	10.38375269 9872	3.461250899 957	1.153750299 986	0.384583433 329
2	(0, -1, -1); (0, 1, 1)	2	3.0042184182 47	1.733268132 242	10.89044486 1908	3.630148287 303	1.210049429 101	0.403349809 700
3	(0, -1, 0); (0, 1, 0)	2	4.9033983123 68	2.214361829 595	13.91324571 2492	4.637748570 831	1.545916190 277	0.515305396 759
4	(0, -1, -2); (0, 1, 2)	2	6.5673810735 49	2.562690202 414	16.10185742 6661	5.367285808 887	1.789095269 629	0.596365089 876
5	(-1, 0, 0); (1, 0, 0)	2	9.2080255294 58	3.034472858 580	19.06615528 0064	6.355385093 355	2.118461697 785	0.706153899 262
6	(-1, 0, -1); (1, 0, 1)	2	10.648820732 769	3.263253090 517	20.50362387 1945	6.834541290 648	2.278180430 216	0.759393476 739
7	(0, 0, -2); (0, 0, 2)	2	10.924685098 847	3.305251140 057	20.76750539 9743	6.922501799 914	2.307500599 971	0.769166866 657
8	(-1, -1, -1); (1, 1, 1)	2	10.924937481 161	3.305289318 828	20.76774528 4040	6.922581761 347	2.307527253 782	0.769175751 261
9	(0, -2, -2); (0, 2, 2)	2	12.016873672 988	3.466536264 485	21.78088972 3816	7.260296574 605	2.420098858 202	0.806699619 401
10	(0, -1, 1); (0, 1, -1)	2	12.264920755 912	3.502130887 890	22.00453733 8610	7.334845779 537	2.444948593 179	0.814982864 393
11	(0, -2, -1); (0, 2, 1)	2	13.084062186 518	3.617189818 978	22.72747392 3882	7.575824641 294	2.525274880 431	0.841758293 477
12	(-1, -1, -2); (1, 1, 2)	2	13.197724065 063	3.632867196 178	22.82597778 9958	7.608659263 319	2.536219754 440	0.845406584 813
13	(-1, 0, 1); (1, 0, -1)	2	13.229572875 569	3.637247981 039	22.85350307 3030	7.617834357 677	2.539278119 226	0.846426039 742
14	(-1, 1, 1); (1, -1, -1)	2	13.499550414 248	3.674173432 794	23.08551252 8960	7.695170842 987	2.565056947 662	0.855018982 554
15	(-1, 1, 0); (1, -1, 0)	2	14.108354236 969	3.756108922 405	23.60032839 3420	7.866776131 140	2.622258710 380	0.874086236 793
16	(-1, -1, 0); (1, 1, 0)	2	14.114493446 682	3.756926063 510	23.60546264 2407	7.868487547 469	2.622829182 490	0.874276394 163
17	(0, -1, -3); (0, 1, 3)	2	15.592886278 275	3.948782885 685	24.81093460 8575	8.270311536 192	2.756770512 064	0.918923504 021
18	(0, -2, -3); (0, 2, 3)	2	16.412027708 881	4.051176089 592	25.45429008 2921	8.484763360 974	2.828254453 658	0.942751484 553
19	(-1, 0, -2); (1, 0, 2)	2	17.551958485 505	4.189505756 710	26.32344101 4903	8.774480338 301	2.924826779 434	0.974942259 811
20	(-1, 1, 2); (1, -1, -2)	2	18.353089140 950	4.284050553 034	26.91748349 0037	8.972494496 679	2.990831498 893	0.996943832 964
21	(-1, -2, -2); (1, 2, 2)	2	18.650286269 358	4.318597720 251	27.13454974 3501	9.044849914 500	3.014949971 500	1.004983323 833
22	(0, -2, 0); (0, 2, 0)	2	19.613593249 472	4.428723659 190	27.82649142 4983	9.275497141 661	3.091832380 554	1.030610793 518
23	(-1, 1, -1); (1, -1, 1)	2	20.179500609 114	4.492159904 669	28.22507311 0516	9.408357703 505	3.136119234 502	1.045373078 167
24	(-1, -1, -3); (1, 1, 3)	2	20.932853198 389	4.575243512 469	28.74710281 4314	9.582367604 771	3.194122534 924	1.064707511 641

Appendix E. Complete 26-Quantity Comparison Ledger

Rank	Quantity	Observed	Nearest declared candidate	Family	Residual ppm
1	electron to shielded helion magn. moment ratio	864.058255	$2^5 * 3^3 = 864$	dyadic-ternary harmonic	67.420
2	deuteron-proton mass ratio	1.99900750127	$2^1 * 3^0 = 2$	dyadic-ternary harmonic	496.496
3	triton-electron mass ratio	5496.92153551	$10^2 \times \text{lambda-sheet count, } g=2 = 5500$	decimal dimensional lift	560.034
4	alpha particle-electron mass ratio	7294.29954171	$10^1 \times \text{Cell3 volume count, } g=2 = 7290$	decimal dimensional lift	589.439
5	helion-electron mass ratio	5495.88527984	$10^2 \times \text{lambda-sheet count, } g=2 = 5500$	decimal dimensional lift	748.691
6	electron g factor	2.00231930436	$2^1 * 3^0 = 2$	dyadic-ternary harmonic	1158.309
7	muon g factor	2.00233184123	$2^1 * 3^0 = 2$	dyadic-ternary harmonic	1164.563
8	neutron-proton mass ratio	1.00137841946	$2^0 * 3^0 = 1$	dyadic-ternary harmonic	1376.522
9	triton-proton mass ratio	2.99371703403	$2^0 * 3^1 = 3$	dyadic-ternary harmonic	2098.717
10	helion-proton mass ratio	2.99315267155	$2^0 * 3^1 = 3$	dyadic-ternary harmonic	2287.664
11	tau-muon mass ratio	16.817	$2^{12} * 3^{-5} = 16.8559670782$	dyadic-ternary harmonic	2317.124
12	neutron-electron mass ratio	1838.683662	$2^{-5} * 3^{10} = 1845.28125$	dyadic-ternary harmonic	3588.213
13	helion g factor	4.2552506995	$2^{-9} * 3^7 = 4.271484375$	dyadic-ternary harmonic	3814.975
14	deuteron-electron mass ratio	3670.48296765	$2^{-4} * 3^{10} = 3690.5625$	dyadic-ternary harmonic	5470.542
15	proton g factor	5.5856946893	$2^{12} * 3^{-6} = 5.61865569273$	dyadic-ternary harmonic	5900.968
16	tau-electron mass ratio	3477.23	$2^7 * 3^3 = 3456$	dyadic-ternary harmonic	6105.434
17	alpha particle-proton mass ratio	3.97259969025	$2^2 * 3^0 = 4$	dyadic-ternary harmonic	6897.325
18	triton g factor	5.95792493	$2^1 * 3^1 = 6$	dyadic-ternary harmonic	7062.034
19	neutron g factor	3.82608552	$2^{-6} * 3^5 = 3.796875$	dyadic-ternary harmonic	7634.571
20	muon-electron mass ratio	206.7682827	$2^{-5} * 3^8 = 205.03125$	dyadic-ternary harmonic	8400.866
21	W to Z mass ratio	0.88145	$2^3 * 3^{-2} = 0.888888888889$	dyadic-ternary harmonic	8439.377
22	electron-neutron magn. moment ratio	960.9205	$2^2 * 3^5 = 972$	dyadic-ternary harmonic	11530.090
23	neutron-muon mass ratio	8.89248408	$2^0 * 3^2 = 9$	dyadic-ternary harmonic	12090.651
24	proton-muon mass ratio	8.88024338	$2^0 * 3^2 = 9$	dyadic-ternary harmonic	13485.736
25	electron to shielded proton magn. moment ratio	658.2275956	$2^3 * 3^4 = 648$	dyadic-ternary harmonic	15538.084
26	deuteron g factor	0.8574382335	$2^{-5} * 3^3 = 0.84375$	dyadic-ternary harmonic	15964.104

Appendix F. Complete Consolidated Analytical Statement

Principal Result: The Mass Bridge Selects The Second Lucas Aggregation

The attached JSON fixes the Cell3 face-ratio point and leaves only uniform scale free. The established Lucas hierarchy defines one Lucas Group as $3 \times 3 \times 3 = 27$ complete Cell3 volumes, with outer basis 3B.

Let the aggregation depth be g ; $m_g = 3^g$, where m_g is the number of Cell3 units along each primitive direction. At $g=0$ there is one Cell3; at $g=1$ there is one Lucas Group; at $g=2$ there is a $3 \times 3 \times 3$ array of Lucas Groups. At $g=2$, $m_2=9$. Nine cells along an axis require ten cell-boundary node positions. Therefore the unrefined corner lattice contains $N_{\text{edge}}=10$, $N_{\text{face}}=10^2=100$, and $N_{\text{volume}}=10^3=1000$. These are exactly the three projective multipliers in the locked Mass Bridge.

Define $r1=A_{\text{ac}}/A_{\text{ab}}=1.339905341992922$, $r2=A_{\text{bc}}/A_{\text{ac}}=1.370359991770000$, and $r3=A_{\text{bc}}/A_{\text{ab}}=1.836152673426000=r1 \cdot r2$. Then the $g=2$ skeleton gives 10 $r1=13.399053419929222=\text{eta}$ alpha, 100 $r2=137.035999177=\text{alpha}^{-1}$, and 1000 $r3=1836.152673426=\text{eta}$.

This forms an exact commutative dimensional closure: $10 \times 100 = 1000$; $(A_{ac}/A_{ab})(A_{bc}/A_{ac}) = A_{bc}/A_{ab}$; and $(\eta \alpha)(\alpha^{-1}) = \eta$. Thus the decimal multipliers are not arbitrary external normalization factors under the recursive-volume interpretation. They are the 1D, 2D, and 3D node cardinalities of the second Lucas aggregation.

Moreover, $3^g 2^n + 1 = 10$ has the unique nonnegative-integer solution $(g,n)=(2,0)$. The Mass Bridge therefore selects a unique hierarchy rung: an unrefined $9 \times 9 \times 9$ Cell3 block containing 27 Lucas Groups and 729 Cell3 volumes.

Accumulated Physical Dimensions

The locked Cell3 has edge lengths $(|a|,|b|,|c|)=(0.3341796791, 0.5864108904, 0.7923197650)$ nm, face areas $(A_{ab},A_{ac},A_{bc})=(0.1948691655, 0.2611062358, 0.3578095391)$ nm², and volume $V_0=0.1179148919$ nm³. Its opposite-face spacings are 0.3295465297, 0.4515973797, and 0.6050977415 nm.

Aggregation preserves the shape exactly: $B_g=3^g B$, $A_g=9^g A_0$, and $V_g=27^g V_0$. The levels are: Cell3 $g=0$, one volume; Lucas Group $g=1$, 27 volumes; Mass-Bridge block $g=2$, 729 volumes; bulk block $g=3$, 19,683 volumes; extended bulk $g=4$, 531,441 volumes. The $g=2$ block has outer edge lengths 3.007617, 5.277698, and 7.130878 nm, volume 85.959956 nm³, outer face areas 15.7844024, 21.1496051, and 28.9825727 nm², opposite-face spans 2.96591877, 4.06437642, and 5.44587967 nm, and body diagonal 12.0928826 nm.

The absolute nanometre dimensions remain conditional on the adopted uniform scale; the ratios, counts, dimensional closure, and reciprocal anisotropy do not.

Growth Of Internal Structure

For the half-step Lucas embedding at aggregation depth g , define $m=3^g$. The merged lattice has $Q_g=(2m+1)^3$ nodes and $S_g=6m+1$ constant-lambda sheets. The central sheet contains $C_g=3m^2+3m+1$ nodes, while the number of antipodal pairs is $P_g=(Q_g-1)/2$.

The hierarchy has the exact sheet-to-volume identity $S_g=Q_{(g+1)}^{(1/3)}$, so the number of sheets at one level becomes the number of half-step node positions along one axis at the next: $7 \rightarrow 7^3=343$, $19 \rightarrow 19^3=6859$, and $55 \rightarrow 55^3=166375$. This is a direct sheet-to-volume recursion. A laminar count at one level becomes a three-dimensional node lattice at the next.

The alternating parity field retains a one-unit global imbalance at every level. Because $m=3^g$ is odd, the sum over all addresses of $(-1)^{(i+j+k)}$ equals -1. The positive and negative populations therefore grow extensively while their difference remains exactly one. The relative imbalance tends to zero, but the absolute residual does not vanish.

Object-To-Field Transition

At $g=0$, Cell3 supplies the local metric, reciprocal metric, three face channels, seven sheets, and 13 antipodal address pairs. At $g=1$, the Lucas Group is the first volume containing a completely surrounded central Cell3. It has 27 translation sectors, 54 internal interfaces, 343 merged half-step nodes, and 19 sheets. Its cell-center adjacency graph has 54 links and 28 independent graph cycles, so it is the first level capable of supporting extensive closed intercell circulation patterns.

At $g=2$, the hierarchy reaches the exact Mass-Bridge closure block. It contains 729 translation sectors, 1,944 internal interfaces, 1,000 unrefined corner nodes, 6,859 half-step nodes, and 55 sheets. More than 71 percent of its half-step nodes are strictly internal, although the strictly interior Cell3 fraction is still slightly below one-half. This is the transition from a boundary-dominated object to a node-bulk-dominated field volume.

At $g=3$, nearly 80 percent of Cell3 volumes and nearly 90 percent of half-step nodes are strictly internal. The internal-to-exposed-interface ratio is exactly 13, equal to the number of local antipodal Cell3 channels. This is the first recursive level that is clearly bulk-dominated in both volume and node populations.

Reciprocal-Space Harmonics

For a periodic block with outer basis mB , where $m=3^g$, the allowed reciprocal vectors are $k_q=2\pi(mB)^{(-T)}q=(2\pi/3^g)B^{(-T)}q$. Using the locked reciprocal metric $R=B^{(-1)}B^{(-T)}$, $|k_q|^2=(2\pi/3^g)^2 q^T R q$.

Consequently, one aggregation step multiplies direct length by 3, area by 9, and volume by 27, while dividing reciprocal wavenumber by 3 and quadratic eigenvalue by 9. One dyadic refinement divides direct length by 2, area by 4, and volume by 8, while multiplying reciprocal wavenumber by 2 and quadratic eigenvalue by 4. The natural linear-dispersion harmonic family is $\omega_{(g,n)}/\omega_0=2^n/3^g$, while quadratic energies or Laplacian eigenvalues scale as $E_{(g,n)}/E_0=4^n/9^g$.

Comparison With Established Physics Constants

A predetermined $2^n 3^g$ harmonic dictionary and canonical hierarchy counts were compared with 26 CODATA dimensionless quantities, excluding the two locked Mass-Bridge anchors from discovery significance. The strongest non-anchor residuals retained for investigation were: electron/shielded-helion magnetic-moment ratio 864.058255 versus $864=2^5 3^3$, residual -67.4 ppm; triton/electron mass ratio 5496.92154 versus $5500=100 \times 55$, residual +560 ppm; alpha/electron mass ratio 7294.29954 versus $7290=10 \times 729$, residual -589 ppm; helion/electron mass ratio 5495.88528 versus 5500, residual +749 ppm; charged-pion/electron mass ratio 273.13244 versus $3^{7/2}/2^3=273.375$, residual +888 ppm; charged-pion/electron mass ratio versus the $g=2$ central sheet count 271, residual -0.781 percent; and tau/electron mass ratio 3477.23 versus the $g=2$ antipodal-pair count 3429, residual -1.387 percent.

The 864.058255 to 864 match is the strongest independent candidate. Under a local log-uniform null it has p approximately 0.0067, but after correction across the 26 tested constants its false-discovery value is approximately $q=0.175$. The $g=2$ nuclear-count candidates have corrected values near $q=0.31$. They warrant retention and residual analysis, but they do not yet constitute independent quantization laws.

A separate dimensionful scan found sub-per-mille approximations to the Planck length, electron Compton wavelength, and Bohr radius using $3^g/2^n$ descendants. Those matches did not survive density correction, and the absolute Cell3 scale remains a free parameter. They therefore cannot presently be used as evidence for a physical scale identification.

Finding

The Mass Bridge does more than force the Cell3 shape. It uniquely selects the $g=2$, $9^3=729$ -Cell3 completion block because that block alone generates the exact 10, 100, and 1000 projective cardinalities. Growing the hierarchy produces deterministic transitions from local geometry, to collective phase sectors, to a Mass-Bridge-complete finite object, and finally to a bulk-dominated field volume. Direct-space aggregation and reciprocal-space mode formation are locked together. Every factor of three in accumulated size produces a factor-of-three spectral subharmonic and 27 times as many collective sectors.

The most important new candidate object is therefore the 9B block: 27 Lucas Groups, 729 Cell3 volumes, 1,000 corner nodes, 6,859 half-step nodes, 55 sheets, and 729 reciprocal phase sectors. It is the minimal recursive volume in which the complete Mass-Bridge dimensional composition law is physically represented by the lattice itself.

The next decisive test is to enumerate the low-order values of $q^T R q$ for the $g=1,2,3$ reciprocal lattices and compare their degeneracies and residual sequences with measured particle, atomic, and field spectra using only one global scale and no nearest-state or object-specific coefficients.

Appendix G. Reproducibility Equations

Face areas: $A_{bc}=|b \times c|$, $A_{ac}=|a \times c|$, $A_{ab}=|a \times b|$.

Node position: $p_L=B L/2$ for L in $\{-1,0,1\}^3$.

Aggregation: $B_g=3^g B$.

Cell count: $N_{\text{cell}}=27^g$.

Unrefined corner nodes: $N_{\text{corner}}=(3^g+1)^3$.

Half-step nodes: $Q_g=(2 \cdot 3^g+1)^3$.

Lambda sheets: $S_g=6 \cdot 3^g+1$.

Central sheet: $C_g=3 \cdot (3^g)^2+3 \cdot 3^g+1$.

Antipodal pairs: $P_g=(Q_g-1)/2$.

Internal interfaces: $I_g = 3(3^g - 1)(3^g)^2$.

Exposed faces: $F_g = 6(3^g)^2$.

Reciprocal modes: $k_q = (2 \pi / 3^g) B^T q$.

Mode quadratic form: $|k_q|^2 = (2 \pi / 3^g)^2 q^T R q$.

Linear harmonic family: $2^n / 3^g$.

Quadratic harmonic family: $4^n / 9^g$.